Practical RIS Gain without Pain via Randomization and Opportunistic Scheduling in 5G NR Systems: Theory and Real-Time Experiments

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Guna Sai Kiran Nekkanti

under the Guidance of

Prof. Debdeep Sarkar and Prof. Chandra R. Murthy

Electrical Communication Engineering
Indian Institute of Science, Bangalore
Bangalore – 560 012 (INDIA)

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Abstract

This work experimentally evaluates the performance of a randomized reconfigurable intelligent surface (RIS) integrated with a real-time fifth-generation (5G) New Radio (NR) system implemented using the OpenAirInterface (OAI) framework. Using an indigenously developed RIS, we first quantify the throughput gains achieved relative to a baseline system without RIS and then examine the impact of randomly sampled RIS phase configurations on key performance metrics, including reference signal received power (RSRP), modulation and coding scheme (MCS) index, block error rate (BLER), and throughput across multiple user equipments (UEs) under proportional fair (PF) scheduling. We further compare the throughput obtained with randomized RIS operation to that achieved using optimized RIS phase configurations commonly considered in the literature. The underlying reason for the observed near-optimal performance is that, in each transmission time slot, the PF scheduler naturally prioritizes the UE experiencing the most favorable instantaneous channel conditions, typically the UE aligned with the randomly configured RIS beam. Additionally, we experimentally demonstrate the tradeoff associated with the RIS switching interval T_s, highlighting how its selection balances channel diversity and stability in practical deployments. Finally, experiments with UEs placed at random locations, both with and without RIS assistance, confirm that randomized RIS consistently enhances system performance, establishing it as a low-overhead and effective alternative to conventional optimization-based RIS approaches in real-world 5G NR systems.

Glossary

Acronym	Definition
3GPP	Third Generation Partnership Project
5G NR	Fifth Generation New Radio
AWGN	Additive White Gaussian Noise
BS	Base Station
BF	Beamforming
BLER	Block Error Rate
BW	Bandwidth
CCDF	Complementary Cumulative Distribution Function
CDF	Cumulative Distribution Function
CLT	Central Limit Theorem
CP	Cyclic Prefix
CS	Cauchy-Scwartz
CSI	Channel State Information
DFT	Discrete Fourier Transform
DL	Downlink
DoA	Direction of Arrival
DoD	Direction of Departure
EE	Energy Efficiency
EWMA	Exponentially Weighted Moving Average
FDD	Frequency Division Duplex
FR	Frequency Range
gNB	Next Generation Node B
HARQ	Hybrid Automatic Repeat Request
HPBW	Half-Power Beam Width
I.I.D.	Independent and Identically Distributed

IRS	Intelligent Reflecting Surface
LoS	Line-of-Sight
MCS	Modulation and Coding Scheme
MIMO	Multiple-Input Multiple-Output
MO	Mobile Operator
MR	Max-Rate
NLoS	Non Line-of-Sight
OC	Opportunistic Communications
OFDM	Orthogonal Frequency Division Multiplexing
PF	Proportional Fairness
QoS	Quality of Service
RF	Radio Frequency
RIS	Reconfigurable Intelligent Surface
RSRP	Reference Signal Received Power
RR	Round Robin
SC	Sub-carrier
SE	Spectral Efficiency
SISO	Single-Input Single-Output
SNR	Signal-to-Noise Ratio
TDD	Time Division Duplex
UE	User-Equipment
UL	Uplink
ULA	Uniform Linear Array
UPA	Uniform Planar Array

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1 | Introduction

1.1 Evolution of Wireless Communication Systems

Wireless cellular communication has undergone continuous evolution since the deployment of first-generation (1G) systems in the 1980s. Each new generation has emerged approximately every decade, driven by the demand for higher data rates, improved reliability, lower latency, and enhanced spectral efficiency. The current fifth-generation (5G) New Radio (NR) standard represents a significant milestone in this evolution, targeting a diverse range of applications across consumer, industrial, and societal domains.

According to the 3GPP vision, 5G NR is designed to support three primary service categories: enhanced mobile broadband (eMBB), massive machine-type communications (mMTC), and ultra-reliable and low-latency communications (URLLC) [1]. These use cases demand unprecedented performance in terms of throughput, latency, connectivity density, and reliability. To meet these requirements, 5G systems employ advanced physical-layer techniques such as massive multiple-input multiple-output (MIMO), operation in millimeter-wave (mmWave) frequency bands, and dense network deployments.

Despite these technological advancements, wireless communication systems remain fundamentally constrained by the uncontrollable nature of the propagation environment. Signal attenuation, blockage, shadowing, and multipath fading continue to limit coverage and reliability, particularly in high-frequency bands. These limitations motivate a paradigm shift in wireless system design, where the propagation environment itself is treated as a controllable entity.

1.2 Scheduling in Multi-User 5G NR Systems

One of the key requirements of 5G New Radio (NR) systems is the support of a large number of users communicating through heterogeneous devices and applications. These conditions give rise to diverse traffic patterns with varying quality-of-service requirements. To efficiently carry such traffic over a wireless medium, the design of scheduling algorithms that adapt resource allocation to user-specific channel conditions is essential. In practice, schedulers are often regarded as the "secret sauce" that determines the performance of a cellular network and play a central role in achieving high throughput, low latency, and fairness among users.

In this thesis, we focus on three widely used scheduling policies that represent different operating points on the throughput–fairness trade-off spectrum: max-rate scheduling, round-robin (RR) scheduling, and proportional fair (PF) scheduling.

1.2.a Role of Scheduling in 5G NR

In 5G NR, scheduling is performed at the medium access control (MAC) layer and determines how time–frequency resources are allocated among multiple user equipments (UEs). By exploiting variations in channel quality across users and time, scheduling enables the system to improve spectral efficiency through intelligent user selection. At the same time, practical schedulers must ensure fairness so that users experiencing unfavorable channel conditions are not permanently starved of resources.

1.2.b Types of Scheduling Policies

1.2.b.i Max-Rate Scheduling

In max-rate scheduling, the gNB selects, in each time slot t, the UE experiencing the strongest instantaneous channel condition. Mathematically, the scheduled user $k^*(t)$ is given by

$$k^*(t) = \arg \max_{k \in \{1,2,\ldots,K\}} |h_k(t)|^2, \tag{1.1}$$

where $h_k(t)$ denotes the channel coefficient between the gNB and the kth UE in a system with K users.

This scheduling policy maximizes the instantaneous system throughput and is optimal from a sum-rate perspective. As the number of UEs increases, the likelihood that at least one user experiences a very strong channel realization also increases. This phenomenon gives rise to a fundamental gain known as *multi-user diversity*.

Multi-User Diversity *Multi-user diversity* refers to the performance gain obtained in a multi-user wireless system by opportunistically scheduling users based on instantaneous channel conditions. In fading environments, different users experience independent channel variations; by selecting the user with the most favorable channel realization in each time–frequency resource, the system achieves an effective increase in the average signal-to-noise ratio (SNR) and spectral efficiency. Importantly, this gain grows with the number of users in the system and is obtained without increasing transmit power or bandwidth.

While max-rate scheduling fully exploits multi-user diversity and achieves the highest possible throughput, it suffers from poor fairness, as users with persistently weak channels (e.g., cell-edge users) are rarely scheduled.

1.2.b.ii Round-Robin Scheduling

Round-robin (RR) scheduling represents the opposite extreme of the scheduling spectrum. In RR scheduling, users are scheduled in a cyclic manner without considering instantaneous channel conditions. This ensures that every UE receives an equal share of resources over time, thereby providing strong fairness guarantees.

A conceptual illustration of RR scheduling in a multi-user system is shown in Fig. 1.1. Since RR scheduling does not exploit channel variations, it fails to harness multi-user diversity and typically results in significantly lower throughput compared to channel-aware schedulers.

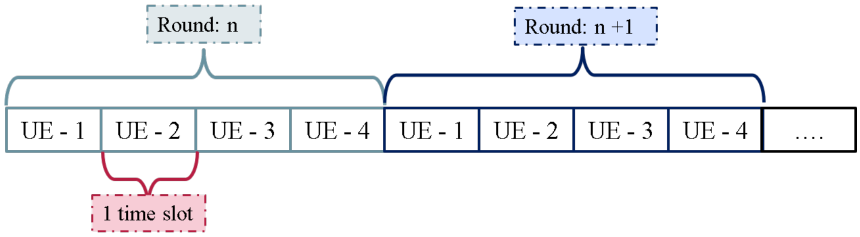


Figure 1.1: Illustration of round-robin (RR) scheduling in a multi-user system.

1.2.b.iii Proportional Fair Scheduling

Proportional fair (PF) scheduling strikes a balance between the throughput optimality of max-rate scheduling and the fairness of RR scheduling. In PF scheduling, the scheduled user $k^*(t)$ in time slot t is selected as

$$k^*(t) = \arg \max_{k \in \{1,2,\ldots,K\}} \frac{R_k(t)}{T_k(t)}, \tag{1.2}$$

where $R_k(t)$ denotes the instantaneous achievable rate of UE k, and $T_k(t)$ denotes its long-term average throughput.

The average throughput $T_k(t)$ is updated recursively as

$$T_k(t+1) = \begin{cases} \left(1 - \frac{1}{T_c}\right) T_k(t) + \frac{1}{T_c} R_k(t), & k = k^*(t), \\ \left(1 - \frac{1}{T_c}\right) T_k(t), & k \neq k^*(t), \end{cases} \tag{1.3}$$

where T_c is the averaging window size that governs the trade-off between throughput and fairness.

Specifically, larger values of T_c make PF scheduling behave closer to max-rate scheduling, thereby emphasizing throughput, while smaller values of T_c make PF scheduling resemble RR scheduling, emphasizing fairness. Due to this flexibility, PF scheduling is widely adopted in practical cellular systems, including 5G NR.

Practical 5G NR implementations, such as the OpenAirInterface (OAI) platform used in this thesis, incorporate scheduling algorithms such as RR and PF to dynamically allocate resources based on observed channel conditions. The presence of such opportunistic scheduling mechanisms is particularly important in the context of RIS-assisted communication, as they provide a natural means to exploit channel fluctuations without requiring explicit physical-layer optimization.

1.3 Intelligent Reflecting Surfaces: Working Principle

Intelligent reflecting surface, also referred to as reconfigurable intelligent surface (RIS), consists of an array of passive elements capable of manipulating an incident electromagnetic wave to steer its reflection in a required direction. A prominent technique to implement an IRS involves the use of metamaterials, which are artificially engineered structures with reconfigurable electromagnetic properties [2]. These materials allow the effective refractive index of each element to be dynamically adjusted, thereby enabling precise control of the reflection coefficient at the point of incidence. Consequently, by tuning the IRS elements to induce specific phase shifts and amplitude responses, it becomes feasible to reconfigure the wireless environment as per our requirement. In particular, this capability facilitates the intelligent control of the wireless channels between any pair of transceivers in the network, opening new avenues for enhancing spectral efficiency (SE), reliable communication, mitigating interference, etc.

To make the discussion more concrete, a simplified schematic of an IRS-assisted wireless communication system is illustrated in Fig. 1.2. This example considers a single BS and a single UE, each equipped with a single antenna. An IRS consisting of N passive reflecting elements is deployed at an appropriate location within the environment, and as described above, the reflection coefficient of each element of the IRS is independently programmable. As a consequence, the overall channel at a UE, say, indexed by k, can be expressed as

$$h_k = \sqrt{\beta_{r,k}}\mathbf{h}_{2,k}^T\mathbf{\Theta}\mathbf{h}_1 + \sqrt{\beta_{d,k}}h_{d,k}, \tag{1.4}$$

where $h_{d,k}$ is the small-scale fading channel from BS to UE-k in the direct link, $\mathbf{h}_1, \mathbf{h}_{2,k} \in \mathbb{C}^N$ are the small-scale fading channels from BS to IRS, and IRS to UE-k respectively, and $\beta_{d,k}, \beta_{r,k}$ denote the large-scale path loss of direct and cascaded channels via the IRS at UE-k. Further, $\mathbf{\Theta} \in \mathbb{C}^{N\times N}$ is a diagonal matrix that contains the reconfigurable reflection coefficients of the IRS elements and is modeled as $\mathbf{\Theta} = \text{diag}\left(\left[\zeta_1 e^{j\phi_1}, \zeta_2 e^{j\phi_2}, \ldots, \zeta_N e^{j\phi_N}\right]^T\right)$, where $\zeta_n \in [0,1]$ and $\phi_m \in [0, 2\pi)$ denote the amplitude and phase of the reflection

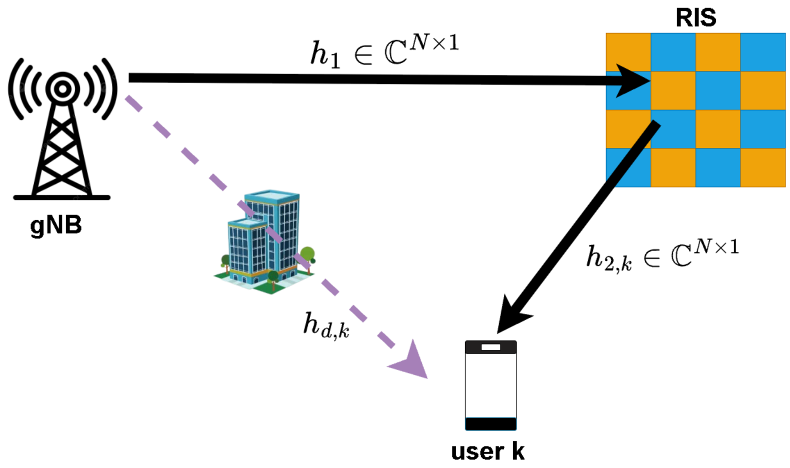


Figure 1.2: An IRS-assisted wireless system.

coefficient at nth IRS element. If we design our IRS circuit elements such that the reflection coefficients have equal magnitude, then, we can set $\zeta_n = 1$ for all $n \in [N] \triangleq \{1, 2, \ldots, N\}$ without loss of generality, and the IRS reflection coefficient matrix, $\mathbf{\Theta}$ becomes a phase shift matrix: $\mathbf{\Theta} = \mathrm{diag}\left(\left[e^{j\phi_1}, e^{j\phi_2}, \ldots, e^{j\phi_N}\right]^T\right)$.

It is evident from (1.4) that the overall wireless channel can be effectively manipulated using the phase-shift array at the IRS, even when both the BS and the UE are equipped with just a single antenna. Extending this idea further, when multiple active antennas are employed at the BS and/or UE in conjunction with a large array of passive reflecting elements at the IRS, the system acquires substantial flexibility. Specifically, the IRS enables us to shape the channel, and the transmitter array at the BS adapts the transmission to channel conditions, thereby allowing for fine-tuned optimization of the end-to-end wireless link in accordance with specific performance objectives.

1.3.a Protocols and Overheads in IRS Integration

We now shift our focus toward understanding the different protocols involved in integrating IRSs into next-generation wireless communication systems. As previously discussed, the IRS modifies the channel by imparting a carefully designed phase shift at each of its elements. However, to apply the desired configuration at the IRS, it is essential to implement a structured *three-phase protocol*, as described below:

1. *Channel estimation*: The first step involves acquiring knowledge of the underlying channel coefficients, also called channel state information (CSI), between the BS and the UE. Specifically, from (1.4), it can be observed that in the presence of an IRS comprising N elements, the total number of channel coefficients is $2N + 1$, including the direct BS–UE path, BS–IRS link, and IRS–UE link. However, since the knowledge of the *cascaded channel*, denoted as $\mathbf{h}_{r,k} \triangleq \mathbf{h}_{2,k} \odot \mathbf{h}_1$, where $\odot$ represents the element-wise (Hadamard) product, is sufficient for optimizing the IRS phase shifts [2], we need to estimate $N + 1$ coefficients, which in turn requires the use of at least $N + 1$ known pilot signals, unless these parameters are significantly correlated.

2. *Channel feedback & phase optimization*:

   (a) The goal of the channel feedback step is to enable the BS to acquire the CSI estimated in the previous step. In the case of frequency division duplexing (FDD), the estimated CSI is fed back to the BS via a dedicated feedback link, after the UE performs channel estimation in the downlink. However, in time-division duplexing (TDD) systems, by leveraging channel reciprocity, the BS can directly estimate the uplink channel and infer the corresponding downlink CSI from it. This, in turn, enables efficient signaling strategies that eliminate the need for explicit CSI feedback from UE to BS, thereby reducing the signaling overheads.

   (b) In the phase optimization step, the BS computes the optimal IRS phase configuration based on the acquired CSI. This typically involves formulating and solving an optimization problem aimed at enhancing a specific performance metric, such as received signal power, spectral efficiency, or signal-to-interference-plus-noise ratio, while satisfying the unit-modulus constraint imposed by the IRS phase shifts.

3. *IRS phase transportation:* This is the final step in configuring an IRS. Owing to the passive nature of the IRS, the BS is responsible for performing the phase optimization process. Once the optimal phase shifts are determined, these phase angles are transported from the BS to the IRS controller through a dedicated control link. The IRS controller then programs each reflecting element with its respective phase shift, thereby

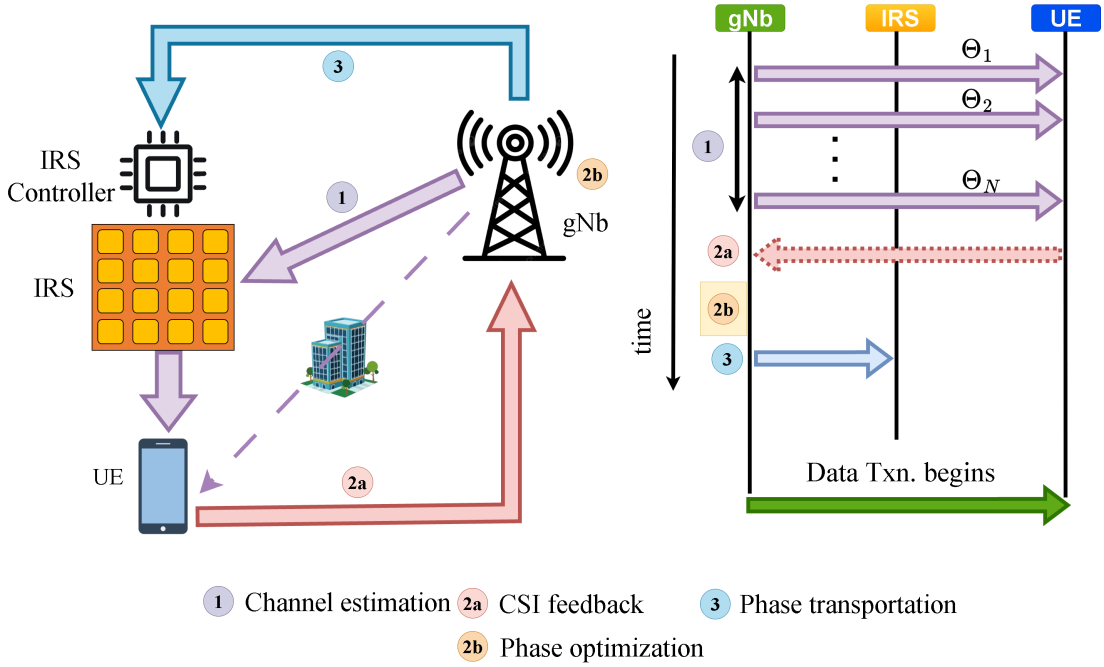


Figure 1.3: Control signaling steps in configuring an IRS. The left-hand figure shows the exchange of information among different nodes, and its timing diagram is shown on the right.

realizing the intended manipulation of the wireless channel.

Figure 1.3 provides a schematic illustration of the complete procedure for configuring an IRS and integrating it into a wireless communication system. These three sources of overhead significantly increase system complexity and can offset the performance gains obtained from IRS deployment, particularly in real-time and fast-varying environments such as 5G NR systems.

1.3.b State-of-the-Art in IRS-Assisted Wireless Communication

Integrating an IRS into a wireless system enables overall channel optimization and can enhance system performance in multiple ways [3–8]. Below, we briefly summarize several approaches proposed in the literature for configuring the IRS and harnessing its benefits:

In [9], the authors demonstrate that an IRS can create a virtual line-of-sight (LoS) path between the BS and UEs, leading to improved SNR and spectral efficiency (SE). Similarly, [10] shows that optimizing IRS phase shifts for a given channel increases the received SNR quadratically with the number of IRS elements. In [11], the authors focus on maximizing the energy efficiency, while works in [12–14] propose joint active and passive beamforming algorithms that underscore the adaptability and controllability of IRS-enabled systems. Moreover, IRS phase optimization in MIMO and orthogonal frequency division multiplexing (OFDM) systems has been addressed in [15–20]. However, realizing the benefits of an IRS requires accurate CSI; accordingly, several works have investigated channel estimation with various guarantees and complexities [21–35]. Coverage enhancement via IRS is explored in [5, 36], the effect of IRSs on inter-BS interference in [37], and IRS-based interference nulling in [38]. Additionally, physical layer security improvements using IRSs are discussed in [39], and the role of IRS in integrated sensing and communications is demonstrated in [40]. These works collectively demonstrate the versatility of IRS technology, but they largely rely on idealized assumptions and optimization-heavy designs.

1.3.c Randomized IRS Configurations: Emerging Directions

A few studies have also investigated randomly configured IRSs. Blind beamforming and diversity analysis with random IRS phases are reported in [41–44], although these methods incur high time complexity. Also, [45] employs randomized IRSs to protect against wireless jammers. For wideband systems, methods for channel estimation and beam training accounting for spatial wideband effects have been developed in [46–48], Localization of UEs is addressed in [49], while [50, 51] leverage wideband phenomena to enhance cell coverage and the performance of OFDM-based multiple access. Finally, several works [52–55] use true time-delay (TTD) units at the IRS to enable coherent beamforming over the entire operational bandwidth.

Although these approaches reduce optimization complexity, they often require long averaging times or large user populations. Moreover, their interaction with practical scheduling mechanisms remains largely unexplored.

1.3.d Experimental Studies on IRS

Regarding deployment scenarios, IRS technology is envisioned for indoor environments, where it can enhance throughput and coverage for IoT applications [56], as well as for outdoor scenarios, as discussed in the previous paragraph.

Beyond academic research, several industrial bodies are also actively advancing through the IRS technology. The European Telecommunications Standards Institute (ETSI) has dedicated study items on IRS [57], the Telecommunications Standards Development Society (TSDSI), an standards development organization of India has proposed interface design options and implementation methods for IRS-aided systems [58, 59], and industry leaders such as Qualcomm and ZTE routinely host workshops, underscoring the growing interest in practical IRS implementation. Experimental validation of IRS-assisted systems is still in its early stages. Prototype-based studies using vector network analyzers and software-defined radios are reported in [60–62]. Coverage enhancement and deployment strategies are explored in [63, 64]. However, these experiments do not consider fully integrated real-time 5G NR stacks.

Only a limited number of works have attempted IRS integration with real-time 5G systems. Most notably, [65] evaluates an IRS in a 5G NR setup but relies on CSI-dependent phase optimization, thereby incurring significant overheads.

1.4 Research Gaps,and Problem Formulation

From the above literature survey, several key research gaps can be identified. First, while reconfigurable intelligent surfaces (RIS) have been shown to provide significant theoretical performance gains, there is a lack of low-complexity RIS operation strategies that are suitable for real-time 5G New Radio (NR) systems. Second, the interaction between RIS configuration timescales and user scheduling dynamics at the medium access control (MAC) layer remains insufficiently understood. Third, there is a scarcity of experimental studies that evaluate RIS-assisted communication using fully integrated real-time 5G NR platforms, where protocol interactions, scheduling decisions, and control overheads are explicitly accounted for.

These gaps are largely attributed to the practical challenges associated with incorporating an RIS into a wireless communication system. In particular, RIS-assisted systems incur the following **three-fold overheads**:

1. **CSI estimation overheads**: Configuring RIS phase coefficients requires knowledge of the cascaded BS–RIS–UE channels. The number of channel coefficients to be estimated scales linearly with the number of RIS elements, resulting in pilot overheads that grow with the RIS size. Even when advanced channel estimation techniques are employed, channel estimation errors persist and typically increase with the number of reflecting elements.

2. **Optimization overheads**: Once CSI is available, determining the optimal RIS phase configuration requires solving a generally non-convex optimization problem. Such optimization procedures are computationally intensive and introduce additional latency, especially for large RIS dimensions.

3. **Control and transportation overheads**: Since RIS elements are passive, the optimized phase coefficients must be conveyed from the base station to the RIS controller via a dedicated control link. The amount of signaling required for this step scales linearly with the number of RIS elements and the phase resolution, further increasing system complexity.

Collectively, these overheads reduce the time and power available for data transmission and can significantly undermine the gains promised by RIS-assisted communication, particularly in real-time 5G NR systems. These observations naturally lead to the following fundamental problem.

Problem Statement

How can one obtain the benefits of an RIS **without explicitly optimizing the RIS** and **without incurring the three-fold overheads** associated with CSI estimation, optimization, and control signaling?

To address this challenge, this thesis adopts a low-complexity system-level approach that leverages mechanisms already present in practical cellular networks.

Proposed Solution

Configure the **RIS phase coefficients randomly** and exploit **opportunistic scheduling** at the base station.

The underlying idea is that randomized RIS phase configurations induce artificial channel fluctuations across users. In a multi-user system, it is highly likely that at least one user equipment (UE) experiences a favorable channel realization for any given RIS configuration. Opportunistic scheduling mechanisms, such as proportional fair (PF) scheduling, can naturally exploit these fluctuations by selecting such UEs for data transmission.

Key Premise

With multiple UEs, a randomly chosen RIS phase configuration is **near-optimal for at least one UE**; by **opportunistically scheduling** that UE, the system captures near-optimal RIS gains *without* explicit RIS optimization.

Motivated by the above problem formulation and solution strategy, this thesis focuses on validating the proposed approach in a *practical 5G NR system*. The specific objectives and contributions of this work are summarized as follows:

- To design and integrate an indigenously built reconfigurable intelligent surface with a real-time 5G New Radio (NR) testbed using the OpenAirInterface (OAI) framework, enabling end-to-end evaluation under realistic protocol constraints.

- To develop a theoretical framework for analyzing randomized RIS-assisted communication combined with opportunistic scheduling, providing insights into the achievable performance gains without explicit RIS optimization.

- To experimentally evaluate the impact of randomized RIS configurations on key system-level performance metrics, including reference signal received power (RSRP), modulation and coding scheme (MCS) adaptation, block error rate (BLER), and user throughput, in a practical 5G NR setting.

- To demonstrate through real-time experiments that randomized RIS operation, when coupled with proportional fair scheduling, can achieve near-optimal performance comparable to optimized RIS schemes, while completely avoiding the CSI estimation, optimization, and control signaling overheads associated with conventional RIS-assisted systems.

1.5 Organization of the Thesis

The remainder of this thesis is organized as follows.

Chapter 2 presents the overall IRS-assisted 5G New Radio (NR) system considered in this work. It introduces the signal and channel models, describes the design and implementation of the indigenously developed reconfigurable intelligent surface (RIS), and explains its integration with a real-time 5G NR testbed based on the OpenAirInterface (OAI) platform, including the scheduling framework.

Chapter 3 develops the theoretical framework for RIS-assisted communication combined with opportunistic scheduling. This chapter analyzes system performance under different scheduling policies and provides insights into the achievable throughput and diversity gains without explicit RIS optimization.

Chapter 4 focuses on experimental validation using a fully integrated real-time 5G NR testbed. It presents the experimental setup and measurement methodology, and evaluates the impact of *randomized and alternate RIS switching* on key performance metrics such as RSRP, MCS adaptation, BLER, and user throughput.

Finally, Chapter 5 concludes the thesis by summarizing the main contributions and outlining directions for future research.

2 | System Description

2.1 Mathematical Description

We consider a base station (BS), referred to as a gNB in 5G New Radio (NR), that serves K user equipments (UEs) using time-division multiple access (TDMA). The communication system is supported by an N-element reconfigurable intelligent surface (RIS) to enhance the quality of the BS–UE links, particularly when direct propagation paths are blocked. The line-of-sight (LOS) link between the BS and UEs is assumed to be completely obstructed, as illustrated in Fig. 2.1.

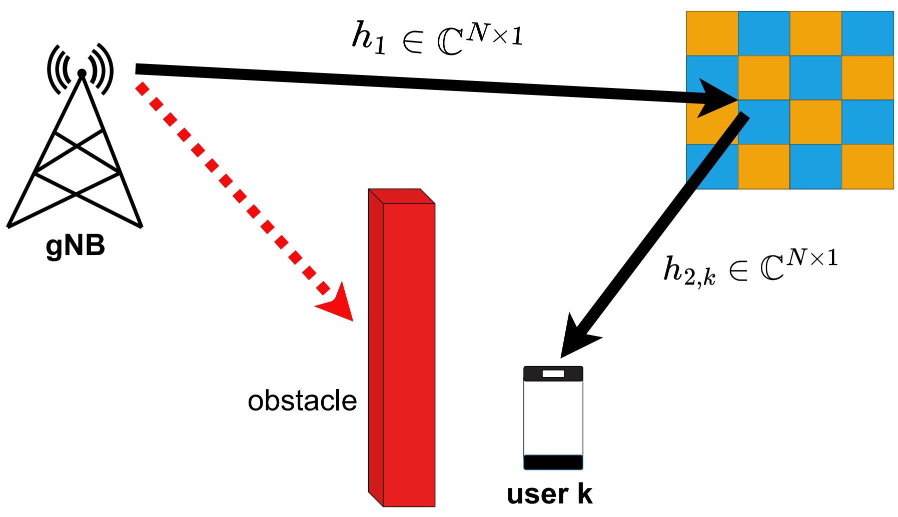


Figure 2.1: A RIS assisted system with LOS blockage.

Let $\mathbf{h}_1 \in \mathbb{C}^N$ denote the BS–RIS channel and $\mathbf{h}_{2,k} \in \mathbb{C}^N$ represent the RIS–UE$_k$ channel. Furthermore, let β_k denote the path loss between the BS and UE$_k$. Accordingly, the received signal at UE_k during time slot t can be expressed as

$$y_k(t) = \sqrt{\beta_k}\sum_{n=1}^{N}[\mathbf{h}_{2,k}]_n[\mathbf{h}_1]_n e^{j\theta_n(t)}x_k(t) + n_k(t) \tag{2.1}$$

$$= \sqrt{\beta_k}\mathbf{h}_{2,k}^T\mathbf{\Phi}(t)\mathbf{h}_1 x_k(t) + n_k(t), \tag{2.2}$$

where $\mathbf{\Phi}(t)$ is a diagonal matrix of RIS phase shifts, defined as

$$\mathbf{\Phi}(t) = \mathrm{diag}\left(e^{j\theta_1(t)}, e^{j\theta_2(t)}, \ldots, e^{j\theta_N(t)}\right),$$

$x_k(t)$ is the transmitted symbol, and $n_k(t)$ denotes the additive white Gaussian noise (AWGN) at UE_k.

To simplify the notation, we define the cascaded BS–RIS–UE channel as

$$\mathbf{h}_{c,k} \triangleq \mathbf{h}_1 \odot \mathbf{h}_{2,k},$$

where $\odot$ denotes the Hadamard (element-wise) product. In addition, let the conjugated RIS phase vector be defined as

$$\boldsymbol{\phi}(t) \triangleq [e^{-j\theta_1(t)}, e^{-j\theta_2(t)}, \ldots, e^{-j\theta_N(t)}]^T.$$

Substituting these definitions into (2.2), the received signal at UE_k can be equivalently written as

$$y_k(t) = \sqrt{\beta_k}\boldsymbol{\phi}^H(t)\mathbf{h}_{c,k}x_k(t) + n_k(t). \tag{2.3}$$

Consequently, the effective channel between the BS and UE_k at time t is expressed as

$$g_k(t) = \sqrt{\beta_k}\boldsymbol{\phi}^H(t)\mathbf{h}_{c,k}. \tag{2.4}$$

This formulation highlights how the RIS dynamically adjusts its phase shifts to optimize the effective channel conditions experienced by each UE. Such flexibility allows the system to compensate for severe propagation losses due to LOS blockage and enhances spectral efficiency through intelligent reflection control.

We next describe the features of the indigenously built RIS and the real-time 5G network

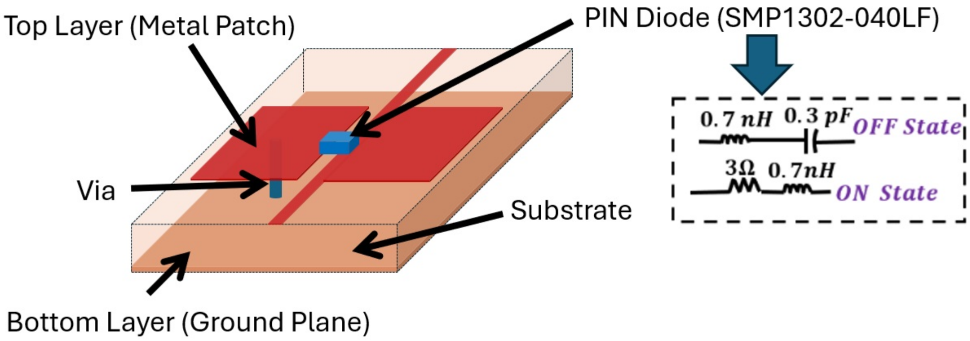


Figure 2.2: Schematic diagram for the unit-cell of 1-bit digitally coded RIS: Perspective 3D view and Equivalent circuit of PIN diode [66].

considered in this work.

2.2 Design and Modeling of the Indigenously Built RIS

Our work employs an indigenously developed one-bit coded reconfigurable intelligent surface (RIS) operating at a center frequency of 5 GHz with a bandwidth of 300 MHz [66]. The RIS features a single-layer architecture comprising 1024 unit cells arranged in a 32×32 uniform planar array (UPA). Each unit cell has a square dimension of $\lambda_0/4$, where $\lambda_0 = 60$ mm is the free-space wavelength at 5 GHz. The compact single-layer structure minimizes insertion loss while maintaining high mechanical stability and ease of fabrication.

Each RIS unit cell integrates an SMP1302–040LF PIN diode to enable reconfiguration between two distinct reflection states (see Fig. 2.2). In the ON state, the diode is modeled as a series connection of a resistor and inductor with values $R_{\text{ON}} = 3\ \Omega$ and $L_s = 0.7$ nH, respectively. In the OFF state, it is represented by a series combination of a capacitor and inductor with $C = 0.3$ pF and $L_s = 0.7$ nH. Electromagnetic simulations using periodic boundary conditions yield reflection phases of -28° (ON) and 150° (OFF), corresponding to a phase difference of approximately 180°. This phase contrast is well-suited for implementing efficient one-bit digital coding, enabling low-complexity beam control. Depending on the illumination and desired beam-steering angles, the required reflection-phase gradient across the RIS surface is analytically computed following [67]. The one-bit quantized phase profile is then realized using discrete reflection phase levels of 0° and 180°

at suitable locations on the UPA. Full-wave electromagnetic simulations in CST Microwave Studio confirm scan-loss-free beam steering over an angular range of 0°–60°, achieving a half-power beamwidth (HPBW) of 6° and sidelobe levels below −9 dB. The simulated results are consistent with theoretical predictions, verifying the reliability of the analytical phase-gradient design.

The phase states of the RIS elements are dynamically controlled through a field-programmable gate array (FPGA)-based control module, which drives the biasing network of PIN diodes. The FPGA allows real-time reconfiguration of the reflection pattern, enabling adaptive beam steering based on the desired transmission direction or user location. To ensure stable operation, proper DC isolation and biasing lines are incorporated into the RIS layout. The beam-steering functionality is experimentally validated in an anechoic chamber setup [66], where the measured radiation patterns closely match simulation results, demonstrating accurate beam directionality and low side-lobe distortion.

The proposed one-bit coded metasurface RIS provides wide-angle beam-steering capability with improved angular resolution compared to conventional reflectarray-based RIS architectures. The analytical phase-distribution approach simplifies codebook generation and reduces computational complexity for real-time operation. Furthermore, by employing PIN diodes and radial stubs instead of varactor diodes and on-chip inductors, the system achieves a significantly lower implementation cost, reduced biasing complexity, and scalability to large array dimensions. The modular and cost-efficient design makes this RIS architecture a strong candidate for future reconfigurable wireless systems operating in sub-6 GHz and mmWave frequency bands.

2.3 RIS-Aided 5G NR via OAI frame work

2.3.a Testbed Architecture

The 5G NR testbed is implemented using the OpenAirInterface (OAI) platform [68,69] in a monolithic stand-alone (SA) configuration that supports the complete 5G RAN protocol stack and core network. Universal Software Radio Peripherals (USRPs) serve as radio units (RUs) to enable over-the-air (OTA) communication between the gNB and the UEs. The OAI 5G RAN and UE software stacks are executed on an Intel i7 desktop and laptops equipped with 12 cores running at 4.9 GHz, 32 GB RAM, and Ubuntu 22.04 LTS with low-latency kernels. For synchronization across all radios, an OctoClock provides a 10 MHz reference and a 1 PPS (pulse-per-second) signal to all B210 devices, including both the gNB and the two UEs.

2.3.b Communication Resource

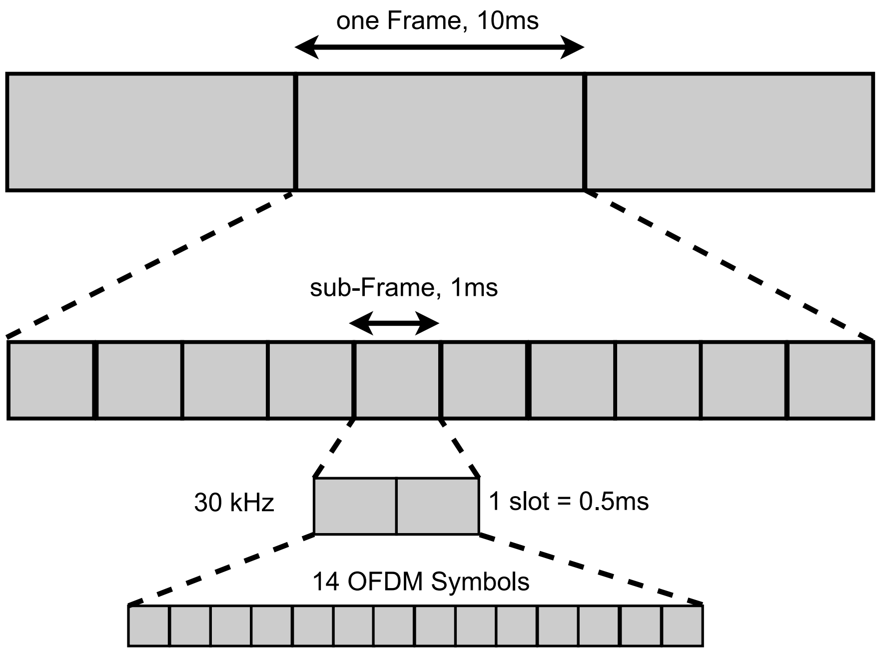


Figure 2.3: System frame structure.

The overall frame structure of our system is illustrated in Fig. 2.3, where frames, sub-frames, slots, and OFDM symbols are nested to highlight the time–frequency organization. The system operates over a 40 MHz bandwidth centered at approximately 5 GHz. With numerology 1 (i.e., 30 kHz subcarrier spacing), each slot has a duration of 0.5 ms and contains 14 orthogonal frequency-division multiplexing (OFDM) symbols.

A time-division duplexing (TDD) scheme with a DL–UL periodicity of 10 slots (5 ms) is employed. Out of every 10 slots, 6 slots are assigned to downlink (DL), 3 slots to uplink

(UL), and 1 slot is configured as a mixed slot comprising 6 DL symbols, 4 guard symbols, and 4 UL symbols. Across the 40 MHz band, the system uses 106 physical resource blocks (PRBs), with approximately 13 symbols per PRB in each slot allocated for scheduling transport blocks [70].

2.3.c Discrete-Rate Adaptation and Scheduling

OpenAirInterface (OAI) employs discrete-rate adaptation to regulate the downlink data rate according to instantaneous channel conditions. The adaptation is governed by the modulation and coding scheme (MCS), which specifies (a) the modulation order and (b) the code rate of the forward error correction (FEC) code. Following the 3GPP-defined 64-QAM MCS Index Table for the physical downlink shared channel (PDSCH) [70], there are 29 possible MCS indices. Higher indices correspond to higher-order constellations and yield improved spectral efficiency (SE). In OAI, MCS selection depends on the maximum tolerable block error rate (BLER) at the user equipment (UE), estimated as a function of the channel quality indicator (CQI) [71]. In our setup, the MCS index ranges from 3 to a CQI-dependent maximum, with CQI updated every 80 ms. Following the hybrid automatic repeat request (HARQ) mechanism [71], the MCS index is increased or decreased by 1 if the measured BLER over a 100 ms window falls below 0.05 or exceeds 0.15, respectively. If a negative acknowledgment (NACK) is received, the same transport block can be retransmitted up to three times using the same MCS before being discarded.

User scheduling in OAI is performed by a proportional-fair (PF) scheduler, which balances throughput maximization with fairness among UEs. At each time slot t, the base station (BS) schedules a single UE, denoted as UE-$k^*(t)$, across all physical resource blocks (PRBs) according to

$$k^*(t) = \underset{k \in \{1,2,\ldots,K\}}{\arg\max}\ \eta_{k,\mathrm{PF}}(t) \triangleq \frac{R_k(t)}{T_k(t)}, \tag{2.5}$$

where $\eta_{k,\mathrm{PF}}(t)$ is the PF metric, $R_k(t)$ is the instantaneous SE, and $T_k(t)$ is the average SE of UE-k. The instantaneous SE is computed using Shannon's formula:

$$R_k(t) = \log_2\left(1 + |g_k(t)|^2 \frac{P}{\sigma^2}\right), \tag{2.6}$$

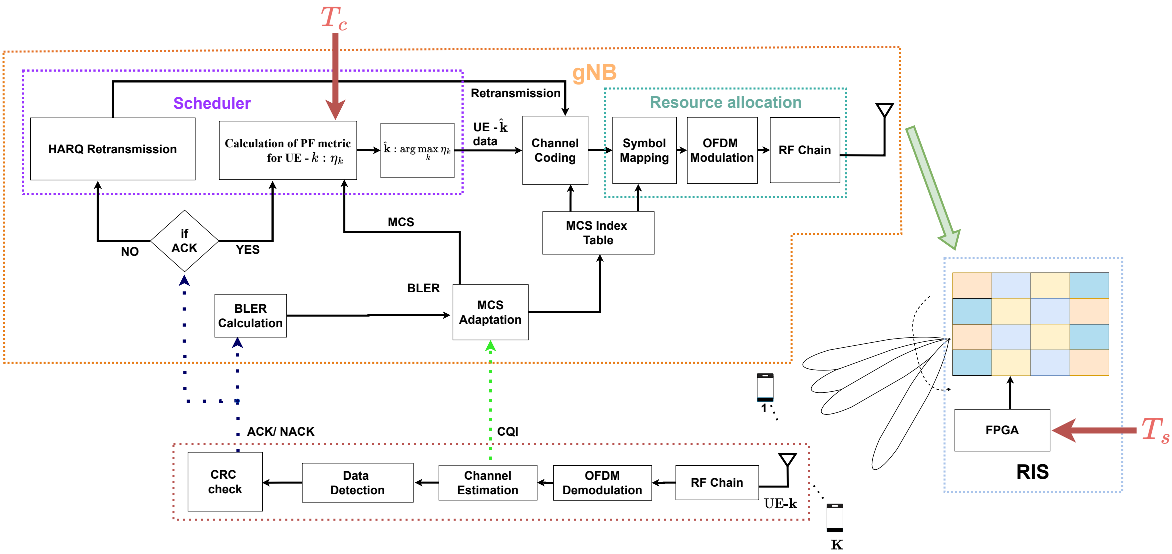


Figure 2.4: Architecture of the proposed RIS-assisted multi-user communication system.

where $g_k(t)$ is the effective channel gain for UE-k, P is the BS transmit power, and σ^2 is the noise variance. The average SE is updated using an exponential-weighted moving average (EWMA):

$$T_k(t+1) = \begin{cases} \left(1 - \dfrac{1}{T_c}\right) T_k(t) + \frac{1}{T_c} R_k(t), & \text{if } k = k^*(t), \\ \left(1 - \dfrac{1}{T_c}\right) T_k(t), & \text{if } k \neq k^*(t), \end{cases} \tag{2.7}$$

where $T_c > 1$ denotes the averaging window. A larger T_c emphasizes long-term throughput (favoring opportunistic scheduling), while a smaller T_c enhances fairness by giving more weight to recent throughput.

2.3.d Overall Mechanism of RIS-aided 5G NR

As shown in the block diagram in Fig. 2.4, the system consists of a gNodeB (gNB), multiple UEs, and an FPGA-controlled RIS. The RIS comprises passive reflecting elements whose phase configurations are controlled by the FPGA. With a switching interval of T_s, the FPGA determines the duration for which a given RIS configuration remains fixed, and the RIS switches to another phase state after every T_s seconds. For a given RIS phase configuration, the downlink communication operates as follows: Each UE continuously monitors the downlink channel quality through channel estimation. Based on this, each UE computes and reports the CQI to the gNB. In addition, the UE that is scheduled for data transmission also performs cyclic redundancy check (CRC) on the decoded data and feeds back ACK/NACK information to the gNB, indicating successful or unsuccessful decoding, respectively. Upon receiving an ACK, the gNB schedules the next UE according to the PF metric defined in (2.5). Then, based on the reported CQI at the scheduled UE, the maximum possible MCS is determined, while the actual MCS used for transmission is adaptively adjusted based on the observed BLER, which is implicitly inferred from HARQ feedback. However, if a NACK was received, the system prioritizes retransmitting the packets to the already scheduled UE using the same MCS under the chase-combining (CC)-based HARQ operation instead of scheduling a new UE []. Subsequently, resource allocation is performed at the gNB, which includes symbol mapping according to the selected MCS and OFDM modulation over a 40 MHz bandwidth. The resulting baseband signal is upconverted to a carrier frequency of 5 GHz and transmitted over the air. The transmitted signal propagates through the RIS-assisted channel and reaches the scheduled UE, where channel estimation, demodulation, decoding, and CRC verification are performed again, and this process continues iteratively. It is important to note that the choice of RIS configuration plays a crucial role in determining the instantaneous channel conditions and, consequently, directly impacts the CQI reports, selected MCS levels, HARQ ACK/NACK outcomes, and the overall throughput and fairness performance across UEs. Finally, since we focus on the impact of RIS on the downlink performance, the CQI and ACK/NACK feedback occurs over a dedicated and reliable uplink control channel.

2.4 Problem Statement

Conventional RIS-based designs rely on explicit optimization of the RIS phase configuration, which requires a structured three-phase protocol and introduces significant overheads. First, *channel estimation* must be performed: in the presence of an RIS with N elements, at least $N+1$ pilot signals are required to estimate the cascaded channel coefficients [2], leading to substantial training overhead. Second, *channel feedback and phase optimization* are necessary: in FDD systems, CSI must be fed back to the BS, while in TDD systems reciprocity can be exploited; in either case, the BS must solve a non-trivial optimization problem to compute the optimal RIS phase configuration. Third, *RIS phase transportation* is required: once the optimal configuration is determined, the BS must transport the phase angles to the RIS controller over a dedicated control link, which then programs each reflecting element. These steps collectively incur training, signaling, and control overheads, limiting the practicality of optimized RIS operation in real-time systems. Motivated by these challenges, our core idea is to replace the explicit RIS optimization step with an approach where the RIS phase configuration is selected randomly and independently over time with an appropriately chosen switching interval T_s between the states. The inherent PF scheduler in 5G NR then naturally prioritizes transmitting data to the UE that experiences the highest instantaneous RSRP under the current RIS configuration, in turn delivering near-optimal RIS benefits without incurring optimization overhead. The objective of this paper is to experimentally evaluate the performance of such a randomized RIS combined with opportunistic PF scheduling in a real-time 5G NR testbed (see Sec. 2.3). Specifically, we investigate:

1. The gain provided by an RIS compared to a system without RIS.
2. The variation of RSRP, MCS index, BLER, and throughput across UEs when RIS configurations are randomly sampled and PF scheduling is applied.
3. The throughput achieved with randomized RIS under PF scheduling compared to optimized RIS phase configurations considered in traditional works.

Furthermore, we demonstrate the tradeoff associated with the switching interval T_s in practical scenarios, showing how its choice balances channel diversity and stability. Finally, by placing users at random locations with and without RIS, we show that randomized RIS consistently enhances performance, thereby validating its effectiveness as a low-overhead alternative to conventional optimization-based approaches.

3 Randomized RIS with Opportunistic scheduling: Theory

3.1 Choice of RIS Configurations

As established in Sec. 2.1, the effective BS–UE$_k$ channel at time t is

$$g_k(t) = \sqrt{\beta_k}\,\boldsymbol{\phi}^H(t)\,\mathbf{h}_{c,k},$$

$$\mathbf{h}_{c,k} = \mathbf{h}_1 \odot \mathbf{h}_{2,k}. \tag{3.1}$$

We now specialize (3.1) to a uniform planar array (UPA) RIS with $N_x \times N_y$ elements ($N = N_x N_y$), inter-element spacings d_x, d_y, and carrier wavelength λ. Let (ν, ψ) denote azimuth and elevation angles (as seen at the RIS). For a UPA, the RIS array response at (ν, ψ) is

$$\mathbf{a}_{\mathrm{RIS}}(\nu,\psi) = \left[e^{j2\pi\left(\frac{d_x}{\lambda} m\cos\psi\cos\nu + \frac{d_y}{\lambda} n\cos\psi\sin\nu\right)} \right]$$
$$m = 0,\ldots,N_x-1;\; n = 0,\ldots,N_y-1.$$

Assume

$$\mathbf{h}_1 = \mathbf{a}_{\mathrm{RIS}}(\nu_{\mathrm{BR}}, \psi_{\mathrm{BR}}),$$
$$\mathbf{h}_{2,k} = \alpha_k\,\mathbf{a}_{\mathrm{RIS}}(\nu_k, \psi_k),$$
$$\alpha_k \overset{\text{i.i.d.}}{\sim} \mathcal{CN}(0,1).$$

where $(\nu_{\mathrm{BR}}, \psi_{\mathrm{BR}})$ are the BS→RIS AoD (azimuth, elevation), and (ν_k, ψ_k) are the RIS→UE$_k$ AoD (azimuth, elevation) at the RIS.

The cascaded channel becomes the Hadamard product of the two UPA responses:

$$\begin{aligned}\mathbf{h}_{c,k} &= \mathbf{h}_1 \odot \mathbf{h}_{2,k} \\ &= \alpha_k \Bigg[e^{j2\pi\left(\frac{d_x}{\lambda} m\left(\cos\psi_{\mathrm{BR}}\cos\nu_{\mathrm{BR}} + \cos\psi_k \cos\nu_k\right)\right.} \\ &\qquad \times e^{j2\pi\left(\frac{d_y}{\lambda} n\left(\cos\psi_{\mathrm{BR}}\sin\nu_{\mathrm{BR}} + \cos\psi_k \sin\nu_k\right)\right)} \Bigg]_{m,n}.\end{aligned}$$

Equivalently, define the effective spatial frequency pair:

$$\begin{aligned}\kappa_x &\triangleq \frac{d_x}{\lambda}\big(\cos\psi_{\mathrm{BR}}\cos\nu_{\mathrm{BR}} + \cos\psi_k\cos\nu_k\big), \\ \kappa_y &\triangleq \frac{d_y}{\lambda}\big(\cos\psi_{\mathrm{BR}}\sin\nu_{\mathrm{BR}} + \cos\psi_k\sin\nu_k\big).\end{aligned}$$

So that

$$\mathbf{h}_{c,k} = \alpha_k \cdot \mathbf{a}_{\mathrm{RIS}}^{\mathrm{eff}}(\kappa_x, \kappa_y), \tag{3.2}$$

$$\left[\mathbf{a}_{\mathrm{RIS}}^{\mathrm{eff}}(\kappa_x, \kappa_y)\right]_{m,n} = e^{j2\pi(m\kappa_x + n\kappa_y)}. \tag{3.3}$$

Substituting (3.3) into (3.1), the effective channel is

$$g_k(t) = \sqrt{\beta_k}\,\alpha_k\,\boldsymbol{\phi}^H(t)\,\mathbf{a}_{\mathrm{RIS}}^{\mathrm{eff}}(\kappa_x, \kappa_y).$$

Let the RIS operate in one of L phase configurations $\boldsymbol{\phi}_1, \dots, \boldsymbol{\phi}_L$, where configuration $\boldsymbol{\phi}_\ell$ is selected with probability p_ℓ, $p_\ell \geq 0$, $\sum_{\ell=1}^{L} p_\ell = 1$. The collection

$$\mathcal{S} \triangleq \Big\{(\boldsymbol{\phi}_1, p_1), \dots, (\boldsymbol{\phi}_L, p_L)\Big\}$$

defines the sampling distribution of RIS configurations.

According to Theorem 1 in [72], the optimal conditional density of the RIS configuration vector corresponding to the cascaded channel $\mathbf{h}_{c,k}$ is

$$g_{\boldsymbol{\theta}}^{\text{opt}}(\boldsymbol{\theta}') = \delta(\boldsymbol{\theta}' - \mathcal{F}(\mathbf{h}_{c,k})), \tag{3.4}$$

where $\mathcal{F}(\cdot)$ denotes the beamforming-optimal mapping

$$\mathcal{F} : \mathbb{C}^N \to \mathcal{C}, \qquad \mathbf{h}_{c,k} \mapsto \exp\{j\,\angle \mathbf{h}_{c,k}\},$$

which maps the cascaded channel vector $\mathbf{h}_{c,k}$ to its corresponding RIS phase configuration $\boldsymbol{\theta}' \in \mathcal{C}$.

Using the law of total probability on (3.4) with respect to the channel distribution, the unconditional density of the RIS configuration variable is

$$f_{\boldsymbol{\theta}}^{\text{opt}}(\boldsymbol{\theta}') = \int \delta(\boldsymbol{\theta}' - \mathcal{F}(\mathbf{h}))p_{\mathbf{h}_c}(\mathbf{h})\, d\mathbf{h}, \tag{3.5}$$

where $p_{\mathbf{h}_c}(\mathbf{h})$ is the PDF of the cascaded channel vector.

For a finite RIS configuration set, the probability mass function (PMF) of configuration $\boldsymbol{\phi}_\ell$ is obtained by evaluating (3.5) at $\boldsymbol{\theta}' = \boldsymbol{\phi}_\ell$:

$$p_\ell = \int \delta(\boldsymbol{\phi}_\ell - \mathcal{F}(\mathbf{h}))\, p_{\mathbf{h}_c}(\mathbf{h})\, d\mathbf{h}, \qquad \ell = 1, \ldots, L. \tag{3.6}$$

Remark 3.1. *Equation* (3.6) *shows that the probability of selecting a given RIS configuration is equal to the measure of the set of channel realizations for which that configuration is beamforming-optimal. Hence, the optimal RIS sampling distribution is entirely determined by the channel statistics and the structure of the mapping* $\mathcal{F}(\cdot)$*, rather than being chosen uniformly or heuristically.*

For the UPA-based RIS, each UE's dominant direction is characterized by its azimuth–elevation pair (ν, ψ), which induces the effective RIS array response $\mathbf{a}_{\text{RIS}}^{\text{eff}}(\kappa_x, \kappa_y)$ defined in (3.3). The beamforming-optimal configuration is

$$\boldsymbol{\phi}^\star(\nu, \psi) = \mathcal{F}(\mathbf{a}_{\text{RIS}}^{\text{eff}}(\kappa_x, \kappa_y)). \tag{3.7}$$

Substituting (3.7) into (3.6), and adopting an empirical angular distribution based on K

users with angles $\{(\nu_k, \psi_k)\}_{k=1}^{K}$, we define

$$f_{\nu,\psi}(\nu, \psi) = \frac{1}{K} \sum_{k=1}^{K} \delta(\nu - \nu_k)\, \delta(\psi - \psi_k),$$

so that the PMF becomes

$$\begin{aligned} p_\ell &= \iint_{\mathcal{A}_\ell} f_{\nu,\psi}(\nu, \psi)\, d\nu\, d\psi \\ &= \frac{1}{K} \sum_{k=1}^{K} \mathbf{1}\{(\nu_k, \psi_k) \in \mathcal{A}_\ell\}, \qquad \ell = 1, \ldots, L, \end{aligned}$$

where $\mathcal{A}_\ell$ is the angular region where $\boldsymbol{\phi}_\ell$ is beamforming-optimal:

$$\mathcal{A}_\ell = \Big\{(\nu, \psi) : \boldsymbol{\phi}_\ell = \arg\max_{\boldsymbol{\phi} \in \mathcal{C}} |\boldsymbol{\phi}^H \mathbf{a}_{\text{RIS}}^{\text{eff}}(\kappa_x, \kappa_y)|\Big\}.$$

Accordingly, the final choice of RIS phase configurations $\{\boldsymbol{\phi}_\ell\}_{\ell=1}^{L}$ is obtained by aligning each configuration to the effective RIS array response corresponding to a representative angular pair $(\nu^{(\ell)}, \psi^{(\ell)})$, i.e.,

$$\begin{aligned} \boldsymbol{\phi}_\ell &= e^{j\,\angle\left(\mathbf{a}_{\text{RIS}}^{\text{eff}}(\kappa_x^{(\ell)}, \kappa_y^{(\ell)})\right)}, \\ \kappa_x^{(\ell)} &= \frac{d_x}{\lambda} \cos\psi^{(\ell)} \cos\nu^{(\ell)}, \\ \kappa_y^{(\ell)} &= \frac{d_y}{\lambda} \cos\psi^{(\ell)} \sin\nu^{(\ell)}, \qquad \ell = 1, \ldots, L. \end{aligned}$$

This construction ensures that each codebook element $\boldsymbol{\phi}_\ell$ is phase-aligned with the effective cascaded channel direction, and the selection probabilities $\{p_\ell\}$ implement the empirical rule:

$$p_\ell = \frac{\text{number of users in } \mathcal{A}_\ell}{\text{total number of users } K}.$$

Remark 3.2. *If the users are uniformly distributed in their azimuth and elevation angles* $(\nu, \psi) \in [-\pi/2, \pi/2]^2$*, then the joint PDF is constant,* $f_{\nu,\psi}(\nu, \psi) = 1/\pi^2$*. In this case, the probability mass function reduces to*

$$p_\ell = \frac{\text{Area}(\mathcal{A}_\ell)}{\pi^2}.$$

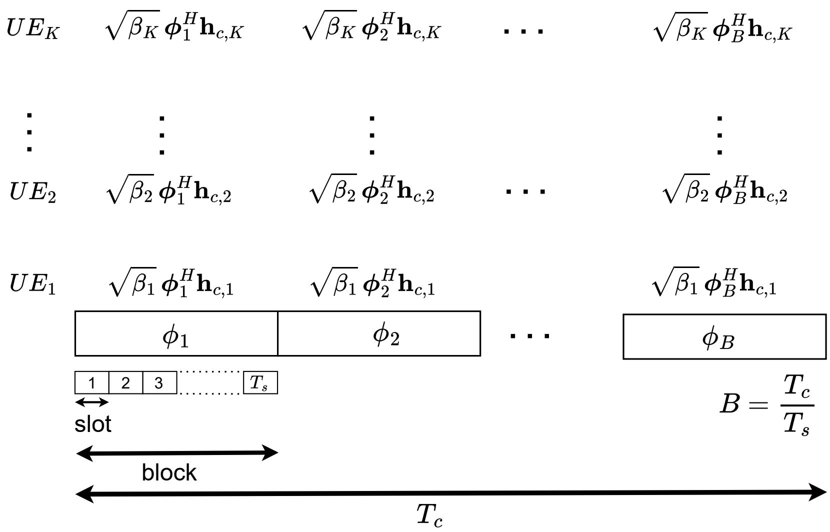


Figure 3.1: Block structure and scheduling. The RIS configuration $\boldsymbol{\phi}_\ell$ remains fixed for T_s time slots, and configurations are selected based on user channel statistics to maximize throughput.

Moreover, if the codebook partitions the angular domain into L regions of equal area, then

$$p_\ell = \frac{1}{L}, \qquad \ell = 1, \ldots, L.$$

Thus, under uniform user distribution and equal-area partitioning, each RIS configuration is selected with equal probability.

The RIS operates in a blockwise manner, maintaining a fixed phase configuration over a block of T_s time slots. At the beginning of each block, the RIS selects one of the L predefined configurations from the set $\mathcal{C}$ and keeps it fixed for the entire block duration, i.e.,

$$\boldsymbol{\phi}(t) = \boldsymbol{\phi}_\ell, \qquad t \in [nT_s, (n+1)T_s - 1],$$

where $\boldsymbol{\phi}(t)$ denotes the active RIS configuration at time slot t, and n is the block index.

3.2 Analysis of Randomized RIS Opportunistic Scheduling

Assumption 3.1. *The instantaneous achievable rate of user k under RIS configuration $\tilde{l}$ is modeled by the linear approximation*

$$R_k^{\tilde{l}}(t) = C_k \, |\boldsymbol{\phi}_{\tilde{l}}^H g_k(t)|^2 \mathrm{SNR}, \tag{3.8}$$

where the effective channel is

$$g_k(t) = \sqrt{\beta_k}\, \alpha_k \, \mathbf{a}_{\mathrm{RIS}}^{\mathrm{eff}}(\kappa_x, \kappa_y), \tag{3.9}$$

with with $\alpha_k \overset{i.i.d.}{\sim} \mathcal{CN}(0,1)$, capturing small-scale fading. $\mathbf{a}_{\mathrm{RIS}}^{\mathrm{eff}}(\kappa_x, \kappa_y)$ denoting the effective RIS array response defined in (3.3).

Theorem 3.1. *Consider a RIS-assisted downlink system with K users and L RIS configurations. The RIS switches independently among these configurations, staying in each for a finite duration of T_s slots. At every switching instant, the configuration is drawn i.i.d. according to the probability mass function $\{p_\ell\}_{\ell=1}^L$ defined in Sec. 3.1.*

Under Assumption 3.1, together with proportional-fair (PF) scheduling and $T_c \to \infty$, the system throughput satisfies

$$\lim_{K\to\infty} \left(R_{\mathrm{sys}}^{(K)} - \frac{1}{K}\sum_{k=1}^{K} R_k^{\mathrm{BF}} \right) = 0, \tag{3.10}$$

where R_k^{BF} denotes the beamforming rate of user k under its optimal RIS configuration.

Proof.

We build upon the RIS configuration construction and sampling distribution $\{p_\ell\}_{\ell=1}^L$ established in Sec. 3.1, where each RIS configuration $\boldsymbol{\phi}_\ell$ is beamforming-optimal for the corresponding angular region $\mathcal{A}_\ell$.

Accordingly, at each switching instant, the RIS operates in configuration $\boldsymbol{\phi}_\ell$ with probability

$$p_\ell = \Pr(\boldsymbol{\phi}(t) = \boldsymbol{\phi}_\ell), \qquad \ell = 1, \ldots, L, \tag{3.11}$$

as induced by the user angular distribution.

Users are classified according to the RIS configuration that maximizes their effective channel gain. The set of users in class L is

$$\mathcal{K}_l = \left\{ k \in \{1, \dots, K\} : \left| \boldsymbol{\phi}_l^H \mathbf{h}_{c,k} \right|^2 = \|\mathbf{h}_{c,k}\|_1^2 \right\}.$$

The empirical fraction of users in class L is

$$\psi_l(K) = \frac{1}{K} \sum_{k=1}^{K} \mathbf{1}\{k \in \mathcal{K}_l\}.$$

By the law of large numbers,

$$\lim_{K \to \infty} \psi_l(K) = p_l, \qquad \forall m,$$

and therefore

$$|\mathcal{K}_l| = \psi_l(K)\, K.$$

At each time slot, the proportional-fair (PF) scheduler selects the user that maximizes the PF metric

$$R_k(t)/T_k(t),$$

where the long-term throughput evolves according to

$$T_k(t+1) = \left(1 - \frac{1}{T_c}\right) T_k(t) + \frac{1}{T_c} R_k(t) \mathbf{1}\{k \text{ scheduled at } t\}.$$

Let the RIS be in configuration $\tilde{l}$. Define

$$Q_{k,l} \triangleq \Pr\left(\frac{R_k^{\tilde{l}}(t)}{T_k(t)} > \frac{R_j^{\tilde{l}}(t)}{T_j(t)},\ \forall j \in \mathcal{K}_l,\ j \neq k \right).$$

The average throughput of user $k \in \mathcal{K}_l$ under the PF scheduler is therefore

$$T_{k,\mathrm{PF}}^{(K)} = \beta_l(K)\, Q_{k,l}\, R_k^{\tilde{l}},$$

where $\beta_l(K)$ is the fraction of time allocated to class $\mathcal{K}_l$. Since $R_k^{\tilde{l}} \le R_k^{\mathrm{BF}}$,

$$T_{k,\mathrm{PF}}^{(K)} \le \beta_l(K)\, Q_{k,l}\, R_k^{\mathrm{BF}} \triangleq T_{k,\mathrm{PF}}^{(K)*}.$$

We now introduce an auxiliary scheduler. Whenever the RIS operates in configuration L, the auxiliary scheduler selects the user in $\mathcal{K}_l$ with the largest PF metric. Hence, at each slot under configuration L, user $k \in \mathcal{K}_l$ is scheduled with probability $Q_{k,l}$.

Over a horizon of T_s slots during which configuration L is active, let N_k denote the number of slots in which user k is scheduled. Then

$$N_k \sim \mathrm{Binomial}(T_s, Q_{k,l}),$$

with mean

$$\mathbb{E}[N_k] = T_s Q_{k,l}.$$

The average throughput of user $k \in \mathcal{K}_l$ under the auxiliary scheduler is therefore

$$\begin{aligned} T_{k,\mathrm{AS}}^{(K)} &= \frac{p_l}{T_s} \mathbb{E}\left[\sum_{t=1}^{N_k} R_k^{\mathrm{BF}}\right] \\ &= \frac{p_l}{T_s}\, \mathbb{E}[N_k]\, R_k^{\mathrm{BF}} \\ &= p_l\, Q_{k,l}\, R_k^{\mathrm{BF}}. \end{aligned}$$

where p_l is defined in (3.11). For $T_c \to \infty$, the PF scheduler maximizes the system utility

$$U_{\mathrm{PF}}(K) = \sum_{k=1}^{K} \log T_k^{(K)}$$

over all feasible schedulers. Using the PF throughput expression and the upper bound, we obtain the sandwich inequality

$$\sum_{k=1}^{K} \log T_{k,\mathrm{AS}}^{(K)} \le \sum_{k=1}^{K} \log T_{k,\mathrm{PF}}^{(K)} \le \sum_{k=1}^{K} \log T_{k,\mathrm{PF}}^{(K)*}.$$

Substituting the expressions for $T_{k,\mathrm{AS}}^{(K)}$ and $T_{k,\mathrm{PF}}^{(K)*}$ and grouping terms by classes yields

$$\sum_{l=1}^{L}\sum_{k\in\mathcal{K}_l}\log\left(p_l Q_{k,l} R_k^{\mathrm{BF}}\right) \leq \sum_{l=1}^{L}\sum_{k\in\mathcal{K}_l}\log\left(\beta_l(K) Q_{k,l} R_k^{\mathrm{BF}}\right).$$

Canceling identical factors gives

$$\sum_{l=1}^{L}|\mathcal{K}_l|\log\left(\frac{p_l}{\beta_l(K)}\right) \leq 0.$$

Dividing by K and using $|\mathcal{K}_l|=\psi_l(K)K$,

$$\sum_{l=1}^{L}\psi_l(K)\log\left(\frac{p_l}{\beta_l(K)}\right) \leq 0.$$

Taking the limit $K\to\infty$ and using $\psi_l(K)\to p_l$ gives

$$\sum_{l=1}^{L}p_l\log\left(\frac{p_l}{\beta_l(K)}\right) \leq 0.$$

The left-hand side is the Kullback–Leibler divergence $D_{\mathrm{KL}}(P\|\beta)$, which is nonnegative and equals zero if and only if $\beta_l(K)=p_l$ for all L. Hence,

$$\lim_{K\to\infty}\beta_l(K)=p_l.$$

Under the effective channel model in (3.9), the instantaneous rate is

$$R_k^{\tilde{l}}(t)=C_k\,\beta_k\,|\alpha_k|^2\,|\boldsymbol{\phi}_{\tilde{l}}^H\mathbf{a}_{\mathrm{RIS}}^{\mathrm{eff}}(\kappa_x,\kappa_y)|^2\,\mathrm{SNR}.$$

By ergodicity of α_k and $\mathbb{E}[|\alpha_k|^2]=1$,

$$T_k(t)=\mathbb{E}[R_k^{\tilde{l}}(t)]=C_k\,\beta_k\,|\boldsymbol{\phi}_{\tilde{l}}^H\mathbf{a}_{\mathrm{RIS}}^{\mathrm{eff}}(\kappa_x,\kappa_y)|^2\,\mathrm{SNR}.$$

Hence the PF metric simplifies to

$$\frac{R_k^{\tilde{l}}(t)}{T_k(t)}=|\alpha_k|^2.$$

Define

$$X_k(t) \triangleq |\alpha_k|^2.$$

Since $\alpha_k \overset{\text{i.i.d.}}{\sim} \mathcal{CN}(0,1)$, we have

$$X_k(t) \sim \text{Exp}(1),$$

and $\{X_j(t)\}_{j \in \mathcal{K}_l}$ are i.i.d. across users. Therefore,

$$Q_{k,l} = \Pr(X_k(t) > X_j(t),\ \forall j \in \mathcal{K}_l,\ j \neq k) = \frac{1}{|\mathcal{K}_l|},$$

by symmetry of continuous i.i.d. exponentials.

Let $K_l = |\mathcal{K}_l|$. Conditioning on $X_k = x$,

$$\begin{aligned} Q_{k,l} &= \int_0^\infty \Pr(X_j < x,\ \forall j \neq k)\, f_{X_k}(x)\, dx \\ &= \int_0^\infty \left(1 - e^{-x}\right)^{K_l - 1} e^{-x}\, dx. \end{aligned}$$

Applying the change of variables $u = 1 - e^{-x}$ yields

$$Q_{k,l} = \int_0^1 u^{K_l - 1}\, du = \frac{1}{K_l} = \frac{1}{\psi_l(K)K}.$$

Substituting back into the asymptotic PF throughput expression and using $\psi_l(K) \to p_l$ as $K \to \infty$, we obtain

$$\lim_{K\to\infty} T_{k,\text{PF}}^{(K)} = \frac{R_k^{\text{BF}}}{K}.$$

Finally, the system throughput is

$$R_{\text{sys}}^{(K)} = \sum_{k=1}^{K} T_{k,\text{PF}}^{(K)},$$

and therefore

$$\lim_{K\to\infty} R_{\text{sys}}^{(K)} = \frac{1}{K}\sum_{k=1}^{K} R_k^{\text{BF}}.$$

This completes the proof. ■

Remark 3.3. *Theorem 1 can be interpreted as follows. As $T_c \to \infty$, the average throughput $T_k(t)$ of each user becomes constant, so the proportional-fair scheduling rule depends only on the instantaneous rate $R_k(t)$. In this regime, the scheduler opportunistically selects the user with the best instantaneous channel. For any finite RIS switching interval T_s, the RIS explores all of its configurations, so each user eventually experiences its favorable channel state. Consequently, the system achieves the full opportunistic gain: for $T_c \to \infty$ and any finite T_s, the throughput converges to the beamforming-optimal benchmark.*

The asymptotic optimality result in Theorem 3.1 relies on the assumption that the proportional-fair (PF) averaging window T_c tends to infinity. This condition guarantees that the empirical probability vectors governing both RIS configuration visitation and PF scheduling decisions converge almost surely to their stationary distributions. In this limit, the system achieves the beamforming-optimal throughput asymptotically, as proven in Theorem 3.1.

In practice, however, T_c cannot be taken arbitrarily large. The parameter T_c serves as the PF averaging horizon used to maintain long-term fairness across users. Scheduling over an infinite horizon would be impractical, since users may not continuously generate data, and the channel statistics may evolve over time. Consequently, a *finite* T_c must be employed, ensuring that the scheduler achieves fairness within each averaging window while still exploiting short-term channel fluctuations for opportunistic gain.

With finite T_c, the achieved throughput over one averaging window becomes a random variable whose deviation from the asymptotic benchmark depends on the accuracy with which empirical distributions track their stationary limits. Specifically, two independent sources of randomness appear:

- **RIS configuration randomness:** Within a window of length T_c, the RIS switches configurations every T_s slots, and thus the empirical visitation frequency vector $\hat{\mathbf{p}} = \{\hat{p}_\ell\}_{\ell=1}^{L}$ may deviate from its stationary probability vector $\mathbf{p} = \{p_\ell\}_{\ell=1}^{L}$.
- **Scheduling randomness:** Even for a fixed RIS configuration, the empirical per-user scheduling distribution $\hat{\mathbf{q}} = \{\hat{q}_k\}_{k=1}^{K}$ under PF scheduling may deviate from its stationary vector $\mathbf{q} = \{q_k\}_{k=1}^{K}$.

These deviations can be quantified using the *total variation (TV) distance*, defined for two probability measures P and Q on a measurable space $(\Omega, \mathcal{F})$ as

$$\|P - Q\|_{\mathrm{TV}} = \sup_{A \in \mathcal{F}} |P(A) - Q(A)|.$$

If P and Q admit densities $p(x)$ and $q(x)$ with respect to a common base measure, this becomes

$$\|P - Q\|_{\mathrm{TV}} = \frac{1}{2} \int |p(x) - q(x)| \, dx,$$

and, for discrete probability vectors $\mathbf{u} = (u_1, \ldots, u_n)$ and $\mathbf{v} = (v_1, \ldots, v_n)$,

$$\|\mathbf{u} - \mathbf{v}\|_{\mathrm{TV}} = \frac{1}{2} \sum_{i=1}^{n} |u_i - v_i|.$$

The TV distance thus measures the maximum discrepancy between two probability laws, quantifying how closely empirical sampling approximates its stationary limit.

Accordingly, we define accuracy parameters $\varepsilon_1, \varepsilon_2 > 0$ satisfying

$$\|\hat{\mathbf{p}} - \mathbf{p}\|_{\mathrm{TV}} \leq \varepsilon_1, \qquad \|\hat{\mathbf{q}} - \mathbf{q}\|_{\mathrm{TV}} \leq \varepsilon_2,$$

where ε_1 controls the accuracy of RIS configuration sampling and ε_2 controls PF scheduling accuracy. The choice of T_c directly determines the magnitudes of these deviations. A sufficiently large averaging window ensures that both empirical distributions approach their stationary distributions with high probability, allowing the finite-time system to closely approximate the asymptotic behavior of Theorem 3.1.

Theorem 3.2. *Consider the RIS-assisted PF scheduling system of Theorem 3.1. Fix accuracy tolerances $\varepsilon_1, \varepsilon_2 \in (0, 2)$ and confidence parameters $\eta_1, \eta_2 \in (0, 1)$. If the PF averaging window T_c satisfies*

$$T_c \geq \max\left\{ T_s \cdot \frac{1}{2\varepsilon_1^2} \ln\left(\frac{2L}{\eta_1}\right), \ \frac{1}{2\varepsilon_2^2} \ln\left(\frac{2K}{\eta_2}\right) \right\}, \tag{3.12}$$

then, with probability at least $1 - \eta_1 - \eta_2$, the PF throughput vector

$$\tilde{\mathbf{T}} \triangleq [\tilde{T}_1, \ldots, \tilde{T}_K]$$

satisfies

$$\|\tilde{\mathbf{T}} - \mathbf{T}^*\|_\infty \leq C_1\varepsilon_1 + C_2\varepsilon_2, \tag{3.13}$$

where $\mathbf{T}^*$ *is the asymptotic throughput vector defined as*

$$\mathbf{T}^* = \frac{1}{K}\,[R_1^{\text{BF}}, R_2^{\text{BF}}, \ldots, R_K^{\text{BF}}]^\top.$$

Proof. Within a PF averaging window of length T_c, the RIS performs T_c/T_s independent configuration switches. Each configuration $\ell \in \{1, \ldots, L\}$ is selected with probability p_ℓ (cf. Sec. 3.1). Denote by X_ℓ the number of times configuration ℓ is selected, and define the empirical visitation frequency

$$\hat{p}_\ell = \frac{X_\ell}{T_c/T_s}.$$

Since $X_\ell \sim \text{Binomial}(T_c/T_s, p_\ell)$, Hoeffding's inequality implies

$$\Pr(|\hat{p}_\ell - p_\ell| > \varepsilon_1) \leq 2\exp\left(-2\frac{T_c}{T_s}\varepsilon_1^2\right).$$

A union bound over all L configurations yields

$$\Pr(\exists \ell:\ |\hat{p}_\ell - p_\ell| > \varepsilon_1) \leq 2L\exp\left(-2\frac{T_c}{T_s}\varepsilon_1^2\right),$$

which is at most η_1 when the first term in the lower bound on T_c holds.

Similarly, define the scheduling indicator $S_k(t) \in \{0, 1\}$, which equals 1 if user k is scheduled at time t. The empirical scheduling frequency is

$$\hat{q}_k(T_c) = \frac{1}{T_c}\sum_{t=1}^{T_c} S_k(t), \qquad q_k = \mathbb{E}[S_k(t)].$$

Applying Hoeffding's inequality again,

$$\Pr(|\hat{q}_k(T_c) - q_k| > \varepsilon_2) \leq 2\exp(-2T_c\varepsilon_2^2),$$

and by the union bound over all K users, this probability is at most η_2 when the second condition in the lower bound on T_c holds.

When both conditions are satisfied, every RIS configuration is visited with frequency

$\hat{p}_\ell = p_\ell \pm \varepsilon_1$, and every user is scheduled with frequency $\hat{q}_k = q_k \pm \varepsilon_2$. The empirical throughput of user k then satisfies

$$\tilde{T}_k(T_c) = \sum_{\ell=1}^{L} \hat{p}_\ell \, \hat{q}_k(T_c) \, R_k^{\mathrm{BF}}.$$

Since the deviations in both empirical distributions are bounded, there exist constants $C_1, C_2 > 0$ such that

$$|\tilde{T}_k(T_c) - T_k^*| \leq C_1 \varepsilon_1 + C_2 \varepsilon_2,$$

with $T_k^* = R_k^{\mathrm{BF}}/K$. This establishes (3.13). ■

Remark 3.4 (Practical Design Implications)**.** *Theorem 3.2 provides a quantitative guideline for selecting a sufficiently large averaging window T_c relative to the RIS switching interval T_s. In practical deployments, T_s should be long enough to accommodate downlink control signaling and coherent data transmission for the currently served UE, yet small enough to allow multiple RIS state transitions within each PF window. Consequently, the ratio T_c/T_s must be large enough to ensure accurate empirical convergence of the RIS visitation and scheduling probability vectors. This requirement corresponds directly to the large-T_c assumption invoked in Theorem 3.1, where asymptotic optimality holds in the limit as $T_c \to \infty$.*

3.3 Simulation Results

This section validates the analytical results presented in Theorems 3.1 and 3.2 through Monte Carlo simulations, demonstrating the performance of the proposed RIS-assisted proportional-fair (PF) scheduling framework under realistic system parameters. A single-antenna base station (BS) is positioned at coordinates $(0, 0)$ m, while a reconfigurable intelligent surface (RIS) comprising 512 reflecting elements is located at $(60, 20)$ m. Ten single-antenna users are uniformly distributed within the rectangular area defined by $x \in [80, 120]$ m and $y \in [-20, 20]$ m. Large-scale path loss is modeled using the standard power-law attenuation with a frequency-dependent constant of $c_0 = -30$ dB. The path loss exponents are set to 2.0 for the BS–RIS link, 2.8 for the RIS–user link, and 3.8 for the

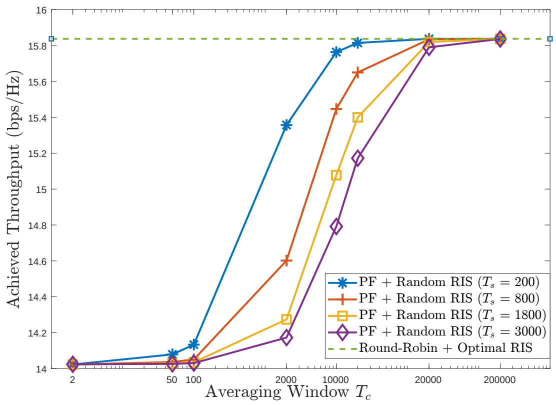


Figure 3.2: System throughput convergence to the beamforming-optimal benchmark with increasing averaging window T_c for different RIS switching intervals T_s.

direct BS–user link. The system operates over a 400 kHz bandwidth at a temperature of 300 K, resulting in a receiver noise power of -117.83 dBm. The BS transmit power is then chosen to ensure an average received signal-to-noise ratio of 110 dB, consistent with the theoretical assumptions. This setting guarantees that all users experience comparable SNR conditions to those analyzed in the performance bounds. Small-scale fading is generated according to independent complex Gaussian coefficients $\alpha_k \sim \mathcal{CN}(0, 1)$ for each user, following the statistical channel model introduced in Sec. 3.1. The RIS employs a codebook of $L = 10$ phase configurations, each aligned toward one of the $K = 10$ users. For the finite-window validation, the accuracy and confidence parameters are set to $\varepsilon = 0.15$ and $\eta = 0.9$, respectively, in accordance with the bounds derived in Theorem 3.2.

Figure 3.2 shows the system throughput as a function of the PF averaging window T_c for several RIS switching intervals T_s, with $K = 10$ users and $L = 10$ RIS configurations. The beamforming-optimal benchmark is obtained from round-robin (RR) scheduling with perfectly phased RIS configurations, representing the upper bound on achievable throughput. As observed, the instantaneous throughput varies with T_s for small T_c due to limited averaging, but all curves converge toward the same benchmark level as T_c increases. This demonstrates that, even with random RIS phase selection, sufficiently large averaging windows allow the PF scheduler to capture full opportunistic gains and approach the beamforming-optimal throughput. The results therefore confirm Theorem 3.1, which

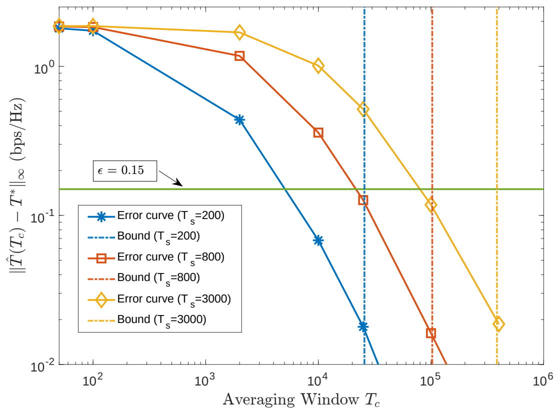


Figure 3.3: Throughput error convergence with PF averaging window T_c for different RIS switching intervals T_s. The solid lines denote simulation results, and dashed lines indicate theoretical bounds.

predicts that for any finite T_s, the system performance converges asymptotically to the optimal value as T_c becomes large, regardless of the RIS switching rate. Figure 3.3 presents the throughput deviation between PF scheduling with the proposed RIS sampling and the RR benchmark as a function of T_c for different values of T_s. The throughput error follows the definition in Theorem 3.2, measuring the maximum deviation between the achieved PF throughput vector and the asymptotic beamforming-optimal reference. The results show that the error decreases monotonically as T_c increases, which is fully consistent with the theoretical prediction. As the averaging window enlarges, the empirical RIS visitation and PF scheduling distributions converge to their stationary probabilities, thus reducing finite-sample variations. Theoretical thresholds $T_c^{\text{th}}(T_s, L, K, \varepsilon, \eta)$ derived from Theorem 3.2 are plotted for comparison. In all cases, the empirical error falls below the target level of $\varepsilon = 0.15$ for T_c values smaller than the theoretical bound, confirming that the analytical guarantees are conservative but accurate at the confidence level $\eta = 0.9$. Overall, the simulation results demonstrate that the proposed RIS-assisted PF scheduling framework achieves excellent agreement with the analytical models. The system throughput converges to the beamforming-optimal limit for large averaging windows, as established by Theorem 3.1, and the finite-time deviations remain tightly bounded in accordance with Theorem 3.2. These observations confirm the robustness and practical validity of the proposed scheduling and RIS configuration design.

4 Randomized RIS with Opportunistic scheduling in 5G NR: Experiments

4.1 Alternate Switching

4.1.a Experimental Setup and Collection of Data

The experimental setup is illustrated in Fig. 4.1. The 5G gNB and UEs are configured as described in Sec. 2.3.a, using horn antennas for transmission and reception. For simplicity, we consider $K = 2$ UEs, and we note that the methodology extends naturally to more users. The UEs are positioned such that $\nu_1 = 30°$, $\nu_2 = 45°$, and $\psi_1 = \psi_2 = 0°$. Based on the discussions in Sec. 3.1, the sampling distribution that maximizes system throughput requires the RIS to alternate between two distinct phase-shift vector states [72, Theorem 1]:

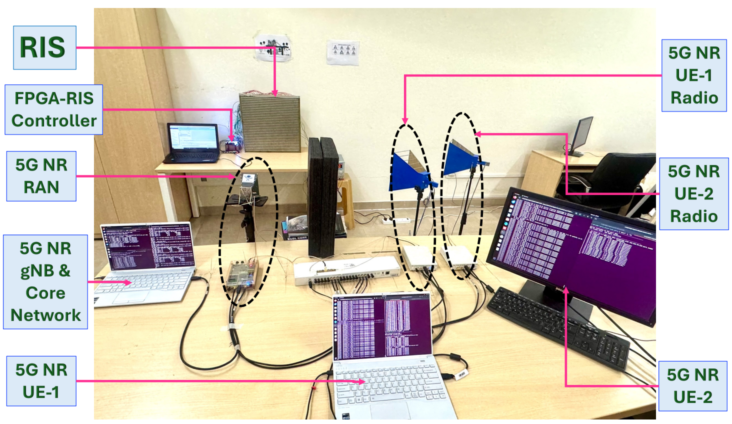


Figure 4.1: Experimental setup used in this work. The system uses an OAI-based 5G NR implementation with 1 gNB and 2 UEs connected to it via an RIS.

$$\boldsymbol{\phi}(t) = \begin{cases} \boldsymbol{\phi}_1 \triangleq \mathbf{a}(\nu_1) \otimes \mathbf{a}(\psi_1), & \text{with probability } \frac{1}{2}, \\ \boldsymbol{\phi}_2 \triangleq \mathbf{a}(\nu_2) \otimes \mathbf{a}(\psi_2), & \text{with probability } \frac{1}{2}, \end{cases} \tag{4.1}$$

where $\otimes$ denotes the Kronecker product, and $\mathbf{a}(\omega)$ represents the array steering vector of an N-element RIS constructed at an angle ω with inter-element spacing $d_{\text{RIS}} = \lambda/4$, given by

$$\mathbf{a}(\omega) = \left[1, e^{-j\frac{\pi}{2}\sin(\omega)}, e^{-j\pi\sin(\omega)}, \ldots, e^{-j(N-1)\frac{\pi}{2}\sin(\omega)}\right]^T. \tag{4.2}$$

From antenna array signal processing theory, setting the RIS phase-shift vector $\boldsymbol{\phi}$ to $\mathbf{a}(\omega)$ directs the reflected signal energy toward the angle ω, i.e., the RIS *beamforms* toward ω.

Remark 4.1. *Although the RIS states in* (4.1) *coincide with those obtained through an optimized design, each state here is chosen independently and with equal probability at every time slot. Consequently, a UE attains optimal beamforming gain only when the randomly selected RIS state aligns with its instantaneous channel. Hence, the RIS is said to be* randomly configured*, indicating that configuring the RIS with any particular phase state is independent of the scheduled UE.*

For real-time collection of data, UEs request downlink data using a transmission control protocol (TCP) over the 5G NR PDSCH. Measurements are collected under two scenarios:

- **Single-UE scenario**: In this case, only one UE is connected to the gNB at a time. For each UE, experiments are conducted both with and without the RIS. When the RIS is present, it is beamformed towards the respective UE under consideration. For all configurations, we record the time traces of RSRP, BLER, MCS, and `iperf` throughput.

- **Scheduling scenario**: Both UEs simultaneously connect and request data from the gNB, which employs a PF scheduler to determine the UE for downlink transmission in each time slot. As a practical realization of the RIS states in (4.1), the RIS alternates between the two states with a carefully chosen *switching time* of $T_s = 9$ seconds, independent of the instantaneous channel conditions (for more details on the choice of the switching time, T_s, please see Remark 4.2 in Sec. 4.1.b.iii). From the perspective of the 5G gNB and UEs, this appears as a random sequence of phase configurations, as explained in Remark 4.1. We record the RSRP, BLER, MCS, and `iperf` throughput at both UEs for $\alpha = 0.01, 0.0005$, and 0.00005.

We next present the main findings, and demonstrate that a randomly configured RIS, when combined with appropriately tuned inherent scheduling mechanisms of a 5G system, can achieve performance comparable to an optimized RIS while avoiding the complexity of real-time phase optimization.

4.1.b Experimental Results: Randomized RIS is Sufficient

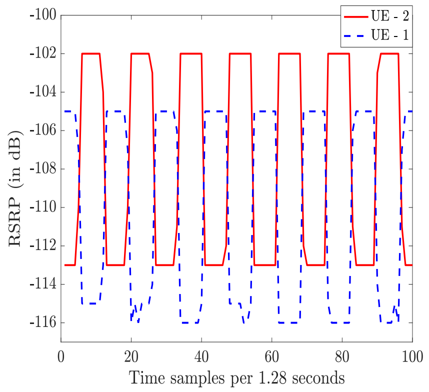
We now report the experimental results obtained from our RIS-aided 5G network using the data collected as described in Sec. 4.1.a. We begin by quantifying the improvements achieved by deploying an RIS and then analyze the temporal behavior of key performance metrics, including RSRP, BLER, MCS, and throughput, under PF scheduling of UEs.

4.1.b.i Quantifying the benefits of the RIS

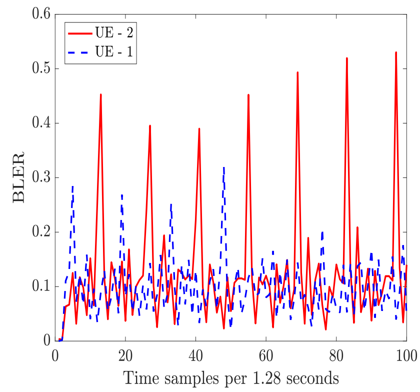
Table 4.1 summarizes the average RSRP values observed at both UEs under the single-UE scenario, with and without the RIS. The RSRP enhances by approximately 7-8 dB, which corresponds to nearly a 7× increase in received signal power, due to the beamforming gain provided by the RIS. Moreover, since the RSRP directly impacts throughput, both UEs experience a noticeable increase in data rates when the RIS is present. These experimental findings, obtained using a physical RIS integrated with a real-time 5G NR system, are in-line with theoretical predictions reported in the literature.

Table 4.1: Performance with and without RIS.(Alternate Switching)

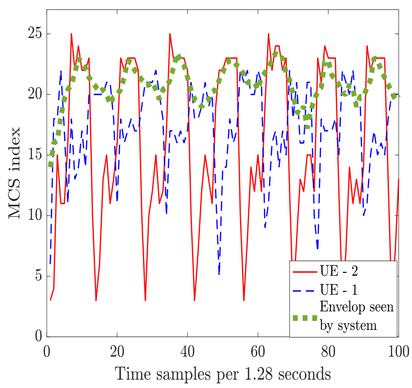
With vs. without RIS	**RSRP (in dB)**		**Throughput (in Mbps)**	
	UE-1	UE-2	UE-1	UE-2
Without RIS	-112	-110	40.70	47.40
With RIS	-105	-102	50.10	58.20



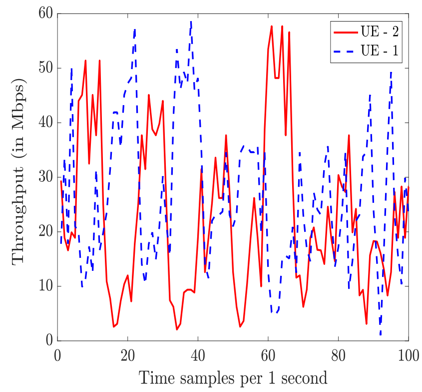
(a) RSRP at two UEs.



(b) BLER at two UEs.



(c) MCS indices for two UEs.



(d) Throughput at two UEs.

Figure 4.2: Variation of performance metrics of a randomized RIS-aided 5G NR system using a proportional-fair scheduler with $\alpha = 0.00005$.

4.1.b.ii Performance of Randomized RIS with PF scheduling

We now examine the variation of performance metrics described in Sec. 4.1.a under the scheduling scenario. Specifically, Fig. 4.2 depicts the temporal evolution of RSRP, BLER, MCS indices, and throughput for both UEs when a PF scheduler operating with $\alpha = 0.00005$ is employed.

Variation of RSRP Fig. 4.2a shows the variation of RSRP at both UEs over time. The RSRP for each UE alternates between high and low levels: -105 dB and -115 dB for UE-1, and -102 dB and -113 dB for UE-2. This periodic oscillation results from the RIS switching its phase configuration between the beamforming directions of the two UEs, as defined in (4.1). So, when one UE experiences a high RSRP, the other UE has a low RSRP, and vice-versa.

Variation of BLER In Fig. 4.2b, we present the variation of BLER experienced by both UEs over time. Recall that a higher BLER indicates that the gNB transmits more information bits than what the channel can reliably support. Accordingly, the MCS is selected such that the BLER at the UEs does not exceed a target value, e.g., 0.1. Thus, the MCS (and the data rate) dynamically adapts to the channel quality. From Fig. 4.2b, it can be observed that, on average, the BLER remains close to the target level of 0.1, with occasional deviations above and below this value. These deviations can be interpreted as follows. When the RSRP at a UE transitions from a low to a high level, the channel quality improves significantly, thereby allowing the UE to support a higher transmission rate. However, since the gNB initially continues to transmit at the lower rate that was used during the low-RSRP period, the UE is able to decode the received data much more reliably than what is desired for, leading to a temporary drop in the BLER below 0.1. Conversely, when the RSRP transitions from high to low, the channel degrades while the gNB continues to transmit at the same higher rate that it used during the high-RSRP phase. This mismatch causes the BLER to increase beyond 0.1 since the transmitted rate now exceeds what the channel can support. Nevertheless, shortly after a transition in RSRP, the gNB adapts its transmission rate to best match the prevailing channel condition, thereby restoring the BLER to its target value of 0.1. This dynamic rate adaptation continues throughout the experiment, ensuring that the BLER is maintained around its desired level.

Variation of MCS Indices Fig. 4.2c illustrates the temporal evolution of the MCS indices at the UEs. As discussed in Sec. 2.3.c, the selection of the MCS index for a given UE depends on the instantaneous channel quality between the gNB and that UE. In an OAI-based 5G NR, the gNB dynamically adapts the MCS based on whether the observed BLER at the scheduled UE is above or below the target value of 0.1. By jointly examining Figs. 4.2b and 4.2c, the following trends can be observed. When the BLER increases beyond 0.1, which typically occurs when the RSRP decreases from a high to a low level, the MCS index correspondingly drops. This reduction lowers the number of information bits transmitted per symbol, thereby improving decoding reliability and bringing the BLER back toward the target level. Similarly, when the BLER falls below 0.1 (i.e., when the RSRP transitions from low to high), the gNB raises the MCS index to exploit the improved channel conditions, thus increasing the data rate while maintaining the desired reliability. During time intervals with no change in RSRP, the MCS index is relatively stable at a value that best matches the prevailing channel quality of the scheduled UE. In Fig. 4.2c, we also show the envelope of the MCS indices, representing the actual MCS levels achieved under the PF scheduler. Notably, despite the RIS configurations being selected randomly, the gNB consistently operates at relatively higher MCS levels. This behavior highlights the effectiveness of the inherent opportunistic scheduling mechanism in 5G systems, which performs data transmission to a UE only when it experiences favorable channel conditions.

Variation of Throughput Fig. 4.2d presents the time evolution of the throughput achieved by the UEs at the application layer. Since a higher MCS index corresponds to transmitting more information bits per symbol, it directly translates to a higher data rate. Accordingly, consistent with the MCS index variation in Fig. 4.2c, we observe that whenever the randomly configured RIS happens to direct its beam toward a given UE, that UE experiences a throughput increase. This occurs because the opportunistic scheduler prioritizes data transmission to the UE currently benefiting from the RIS gain, thereby maximizing the system's instantaneous throughput. This is precisely the well-known *multi-user diversity* effect, wherein the scheduler leverages the channel fluctuations across users to enhance overall performance. Consequently, as will be demonstrated in Fig. 4.3, this adaptive scheduling mechanism enables the system to achieve a substantially higher aggregate throughput, comparable to that obtained via optimized RIS.

4.1.b.iii Randomized RIS Can Procure Full Benefits

In this final sub-section, we show that a randomized RIS, when paired with an opportunistic scheduler, can achieve nearly the maximal throughput associated with optimized RIS configuration. Fig. 4.3 depicts the system throughput versus $1/\alpha$, where α is the EWMA parameter in (2.7). The topmost curve depicts the throughput under round-robin (RR) scheduling of UEs with the RIS optimally configured toward the scheduled UE in each

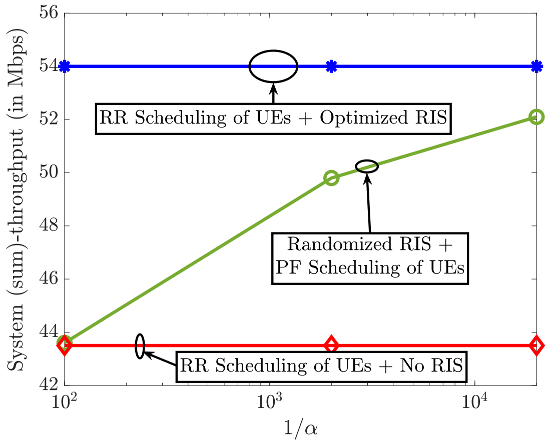


Figure 4.3: System throughput vs. $1/\alpha$.

slot. This genie-aided curve is obtained by averaging the single-UE throughput values with RIS beamforming to the considered UE (i.e., it is computed from Table 4.1); it is independent of α and represents the maximum achievable throughput with an optimized RIS. Practically, achieving this throughput requires tight control of the RIS, configuring it based on the scheduled UE. When the RIS instead switches randomly according to (4.1) and the gNB employs PF scheduling, and α is large, the scheduler prioritizes fairness and allocates resources without accounting for the UEs' instantaneous channels, yielding a relatively low throughput. As α decreases, the scheduler increasingly favors selecting the UE with better instantaneous channels, i.e., the UE toward which the randomly chosen RIS beam is directed, leading the system's performance approaching that of the (genie-aided) optimized RIS [73]. *Importantly, the figure demonstrates that the throughput achieved with a randomized RIS and a properly tuned PF scheduler at the gNB not only consistently surpasses that of the no-RIS case, but can also deliver the full benefits of an optimized RIS even without explicitly optimizing the RIS.*

Remark 4.2 (Choice of α and T_s)**.** *In general, a judicious selection of the RIS switching interval, T_s, is also crucial. Specifically, T_s should be long enough to allow the 5G NR control signaling and scheduling mechanisms to operate and transmit data to the UE currently benefiting from the near-beamforming RIS gain, yet not so long that the PF scheduler (with a given EWMA parameter α) is compelled to serve the other UE to which the RIS is not aligned. Therefore, careful tuning of the (α, T_s) pair is essential to the success of*

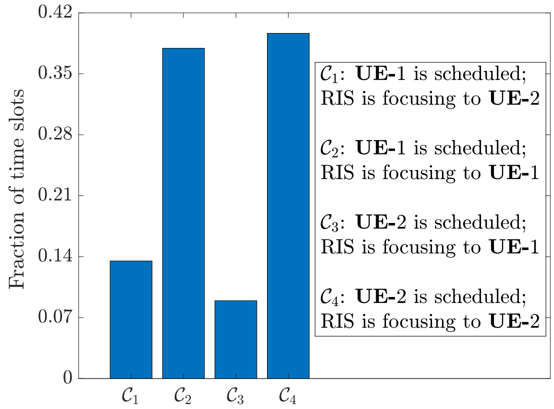


Figure 4.4: Fraction of time slots allotted by PF scheduler with $\alpha = 0.00005$.

the proposed approach. In this work, T_s was empirically chosen to reveal the benefits of randomized RIS operation; a formal analysis of their interdependence is left for future work.

To further substantiate the above findings, Fig. 4.4 presents a histogram showing the fraction of time slots in which each UE is scheduled during its respective favorable (good) and unfavorable (bad) states, based on a single scheduling instance with $\alpha = 0.00005$ observed over a long time trace of channel realizations. Specifically, the figure shows the proportion of time slots in which a PF scheduler selects a UE for transmission when the RIS is aligned with that UE's channel versus when it is not. We see that, despite the randomized nature of the RIS configurations, the scheduler predominantly transmits data to a UE when the RIS is in the beamforming configuration favoring that UE. Occasional instances occur where a UE is scheduled even when the RIS is not aligned toward it, for example, when packets need to be retransmitted (in 5G NR, retransmissions are prioritized above PF scheduling decisions.)

Moreover, the PF scheduler also ensures that the UEs are scheduled almost equally over time, each receiving about 40% of the total time slots during which the RIS focuses its energy towards them, thereby maintaining fairness across UEs.

4.2 Random Switching

4.2.a Experimental Setup and Collection of Data

The experimental setup is illustrated in Fig. 4.5. The 5G gNB and UEs are implemented using the OpenAirInterface (OAI) framework as described in Sec. 2.3.a, with Universal Software Radio Peripheral (USRP) devices serving as the radio front-ends. Although the USRPs include built-in antennas for over-the-air transmission and reception, external horn antennas were employed to provide additional directional gain and improve link reliability during measurements. For simplicity, we consider $K = 2$ UEs, and we note that the methodology extends naturally to a larger number of users. The UEs are positioned such that $\nu_1 = 30°$, $\nu_2 = 60°$, and $\psi_1 = \psi_2 = 0°$. Following the discussion in Sec. 3.1, the throughput-maximizing sampling distribution requires the RIS to operate across two representative phase-shift states [72, Theorem 1], which correspond to the dominant angular directions of the two UEs. Specifically, the RIS phase configuration at time t is selected as

$$\boldsymbol{\phi}(t) = \begin{cases} \boldsymbol{\phi}_1 \triangleq \mathbf{a}(\nu_1) \otimes \mathbf{a}(\psi_1), & \text{with probability } \frac{1}{2}, \\ \boldsymbol{\phi}_2 \triangleq \mathbf{a}(\nu_2) \otimes \mathbf{a}(\psi_2), & \text{with probability } \frac{1}{2}, \end{cases} \tag{4.3}$$

where $\otimes$ denotes the Kronecker product, and $\mathbf{a}(\omega)$ represents the array steering vector of an N-element RIS constructed for an angle ω with inter-element spacing $d_{\mathrm{RIS}} = \lambda/4$, given by

$$\mathbf{a}(\omega) = \left[1,\, e^{-j\frac{\pi}{2}\sin(\omega)},\, e^{-j\pi\sin(\omega)},\, \ldots,\, e^{-j(N-1)\frac{\pi}{2}\sin(\omega)}\right]^T. \tag{4.4}$$

In this experiment, the RIS randomly switches between the two phase configurations $\boldsymbol{\phi}_1$ and $\boldsymbol{\phi}_2$ at each time slot, each with equal probability. This random switching emulates the statistically optimal RIS operation derived in Theorem 1, ensuring that the RIS reflects signals toward both users over time in proportion to their channel occurrence probabilities, without explicitly coupling the RIS state to the instantaneous scheduling decision. For real-time data collection, the UEs request downlink traffic using the Transmission Control Protocol (TCP) over the 5G NR PDSCH. Measurements are collected under two experimental scenarios:

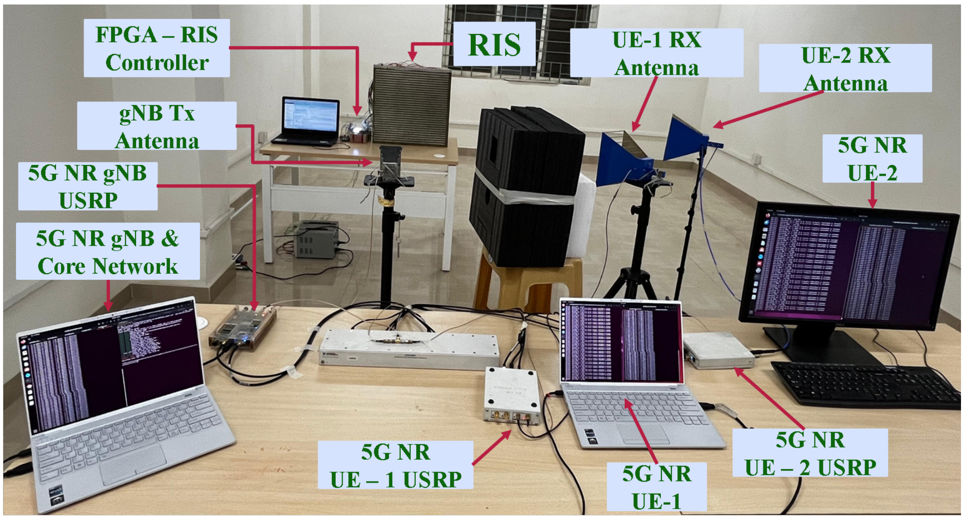


Figure 4.5: Experimental setup used in this work. The system uses an OAI-based 5G NR implementation with 1 gNB and 2 UEs connected to it via an RIS.

- **Single-UE scenario**: Only one UE is connected to the gNB at a time. For each UE, experiments are conducted both with and without the RIS. When the RIS is active, it is beamformed toward the corresponding UE under consideration. In all configurations, time traces of the reference signal received power (RSRP), block error rate (BLER), modulation and coding scheme (MCS), and `iperf` throughput are recorded.

- **Scheduling scenario**: Both UEs are simultaneously connected and continuously request downlink data from the gNB, which employs a proportional-fair (PF) scheduler to select the transmitting UE at each time slot. In this setup, the RIS switches *randomly* between the two phase configurations defined in (4.3), with switching intervals of $T_s = 3$, 5, and 15 seconds. The random selection of RIS states is independent of the instantaneous channel conditions and emulates the statistical RIS operation described in Sec. 3.1. For each experiment, the RSRP, BLER, MCS, and `iperf` throughput are recorded at both UEs for PF averaging window lengths corresponding to $T_c = 200$, 2000, and 20000 slots.

MCS adaptation

As outlined in Sec. 2.3.c, OpenAirInterface (OAI) performs MCS adaptation using CQI feedback and the 3GPP-defined MCS tables [71]. This approach works well in MIMO systems, but it introduces a limitation in single-input single-output (SISO) setups. According to the 3GPP specifications, CQI reporting is not defined for SISO configurations [70]. In practice, OAI therefore assumes a high CQI value by default and MCS adaptation happens through only by BLER, which leads to slow variation of MCS according to BLER thresholds and this is not efficient MCS adaptation in RIS-assisted communication systems where real channel conditions vary rapidly due to RIS switching.

To address this issue, we modified the OAI code to replace CQI-based MCS adaptation with an RSRP-based mechanism. Since RSRP is consistently reported in our setup, we use it to determine the maximum MCS index as follows:

- If RSRP > -102 dBm, set $\text{MCS}_{\text{max}} = 25$,
- If RSRP < -115 dBm, set $\text{MCS}_{\text{max}} = 10$,
- Otherwise, set $\text{MCS}_{\text{max}} = 15$.

These thresholds were empirically selected to balance efficiency and reliability. Stronger RSRP values permit higher-order modulation and coding, while weaker RSRP values restrict the maximum MCS to maintain robustness. After setting the maximum MCS index based on RSRP, the adaptation continues to follow the BLER-driven adjustment mechanism, where the MCS index is increased or decreased depending on observed BLER thresholds. This modification ensures that scheduling decisions are realistic and better aligned with actual channel conditions in SISO deployments, leading to improved throughput convergence and system stability.

We next present the main findings, and demonstrate that a randomly configured RIS, when combined with appropriately tuned inherent scheduling mechanisms of a 5G system, can achieve performance comparable to an optimized RIS while avoiding the complexity of real-time phase optimization.

4.2.b Experimental Results: Randomized RIS is Sufficient

We now report the experimental results obtained from our RIS-aided 5G network using the data collected as described in Sec. 4.2.a. We begin by quantifying the improvements achieved by deploying an RIS and then analyze the temporal behavior of key performance metrics, including RSRP, BLER, MCS, and throughput, under PF scheduling of UEs.

4.2.b.i Quantifying the benefits of the RIS

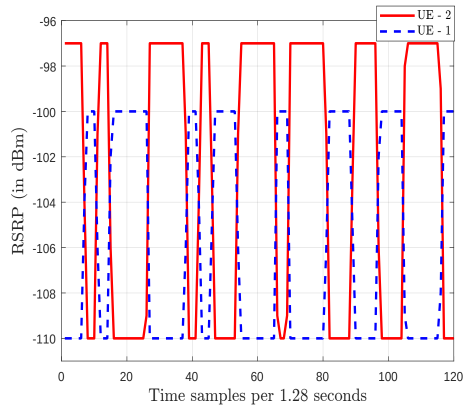
Table 4.2: Performance with and without RIS.

With vs. without RIS	**RSRP (in dBm)**		**Throughput (in Mbps)**	
	UE-1	UE-2	UE-1	UE-2
Without RIS	-110	-110	25.2	26.3
With RIS	-100	-97	51.1	55.9

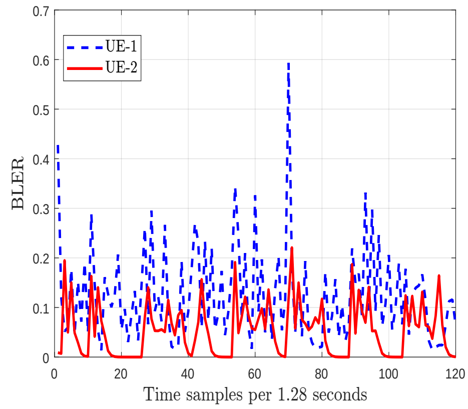
Table 4.2 summarizes the average RSRP values observed at both UEs under the single-UE scenario, with and without the RIS. The RSRP enhances by approximately 10-13 dB, which corresponds to nearly a $10\times$ increase in received signal power, due to the beamforming gain provided by the RIS. Moreover, since the RSRP directly impacts throughput, both UEs experience a noticeable increase in data rates when the RIS is present. These experimental findings, obtained using a physical RIS integrated with a real-time 5G NR system, are in-line with theoretical predictions reported in the literature.

4.2.b.ii Performance of Randomized RIS with PF scheduling

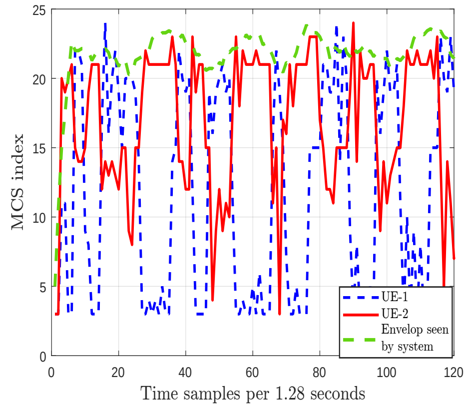
We now examine the variation of performance metrics described in Sec. 4.2.a under the scheduling scenario. Specifically, Fig. 4.6 illustrates the temporal evolution of RSRP, BLER, MCS index, and throughput for both UEs when a proportional-fair (PF) scheduler with an averaging window of $T_c = 20000$ is employed. The RIS operates with a switching interval of $T_s = 5$ seconds, which was identified as the most effective configuration during experimentation. The rationale behind selecting this switching duration, along with the observed trade-offs associated with different T_s values, is discussed in Sec. 4.2.b.iii.



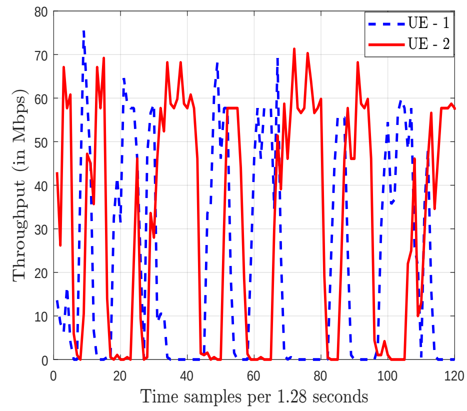
(a) RSRP at two UEs.



(b) BLER at two UEs.



(c) MCS indices for two UEs.



(d) Throughput at two UEs.

Figure 4.6: Variation of performance metrics of a randomized RIS-aided 5G NR system using a proportional-fair scheduler with $T_c = 20000$.

Variation of RSRP Fig. 4.6a shows the variation of RSRP at both UEs over time. The RSRP for each UE alternates between high and low levels: -100 dBm and -110 dBm for UE-1, and -97 dBm and -110 dBm for UE-2. This behavior results from the RIS switching its phase configuration between the beamforming directions of the two UEs, as defined in (4.3). Since the RIS chooses its configurations randomly, the durations for which each UE experiences high or low RSRP vary, leading to different duty cycle lengths. Thus, when one UE experiences a high RSRP, the other UE has a low RSRP, and vice-versa.

Variation of BLER In Fig. 4.6b, we present the variation of BLER experienced by both UEs over time. Recall that a higher BLER indicates that the gNB transmits more information bits than what the channel can reliably support. Accordingly, the MCS is selected such that the BLER at the UEs does not exceed a target value, e.g., 0.1. Thus, the MCS (and the data rate) dynamically adapts to the channel quality.

From Fig. 4.6b, it can be observed that, on average, the BLER remains close to the target level of 0.1, with occasional deviations above and below this value. These deviations can be interpreted as follows. When the RSRP at a UE transitions from a low to a high level, the channel quality improves significantly, thereby allowing the UE to support a higher transmission rate. However, since the gNB initially continues to transmit at the lower rate that was used during the low-RSRP period, the UE is able to decode the received data much more reliably than what is desired for, leading to a temporary drop in the BLER below 0.1. Conversely, when the RSRP transitions from high to low, the channel degrades while the gNB continues to transmit at the same higher rate that it used during the high-RSRP phase. This mismatch causes the BLER to increase beyond 0.1 since the transmitted rate now exceeds what the channel can support. Nevertheless, shortly after a transition in RSRP, the gNB adapts its transmission rate to best match the prevailing channel condition, thereby restoring the BLER to its target value of 0.1. This dynamic rate adaptation continues throughout the experiment, ensuring that the BLER is maintained around its desired level.

Variation of MCS Indices Fig. 4.6c illustrates the temporal evolution of the MCS indices at the UEs. As discussed in Sec. 4.2.a, the selection of the MCS index for a given UE depends on the instantaneous channel quality (RSRP) between the gNB and that UE. In an OAI-based 5G NR, the gNB dynamically adapts the MCS based on whether the observed BLER at the scheduled UE is above or below the target value of 0.1.

By jointly examining Figs. 4.6b and 4.6c, the following trends can be observed. When the BLER increases beyond 0.1, which typically occurs when the RSRP decreases from a high to a low level, the MCS index correspondingly drops. This reduction lowers the number of information bits transmitted per symbol, thereby improving decoding reliability and bringing the BLER back toward the target level. Similarly, when the BLER falls below 0.1 (i.e., when the RSRP transitions from low to high), the gNB raises the MCS index to exploit the improved channel conditions, thus increasing the data rate while maintaining the desired reliability. During time intervals with no change in RSRP, the MCS index is relatively stable at a value that best matches the prevailing channel quality of the scheduled UE.

In Fig. 4.6c, we also show the envelope of the MCS indices, representing the actual MCS levels achieved under the PF scheduler. Notably, despite the RIS configurations being selected randomly, the gNB consistently operates at relatively higher MCS levels. This behavior highlights the effectiveness of the inherent opportunistic scheduling mechanism in 5G systems, which performs data transmission to a UE only when it experiences favorable channel conditions.

Variation of Throughput Fig. 4.6d presents the time evolution of the throughput achieved by the UEs at the application layer. Since a higher MCS index corresponds to transmitting more information bits per symbol, it directly translates to a higher data rate. Accordingly, consistent with the MCS index variation in Fig. 4.6c, we observe that whenever the randomly configured RIS happens to direct its beam toward a given UE, that UE experiences a throughput increase. This occurs because the opportunistic scheduler prioritizes data transmission to the UE currently benefiting from the RIS gain, thereby maximizing the system's instantaneous throughput. This is precisely the well-known *multi-user diversity* effect, wherein the scheduler leverages the channel fluctuations across users to enhance overall performance. Consequently, as will be demonstrated in Fig. 4.7, this adaptive scheduling mechanism enables the system to achieve a substantially higher aggregate throughput, comparable to that obtained via optimized RIS.

4.2.b.iii Tradeoff between T_s and T_c

As discussed in Remark 3.4, the RIS switching interval T_s plays a critical role in determining the accuracy of throughput for a given PF averaging window T_c. To investigate this tradeoff, we conducted experiments with three different switching times, $T_s = 3$ s, 5 s, and 15 s, while varying T_c across representative values. The results reveal a consistent pattern, summarized in Table 4.3. When T_s is too small, the frequent RIS reconfigurations lead to reduced throughput due to modulation and coding scheme (MCS) adaptation overheads and control signaling inefficiencies. Conversely, when T_s is too large, the PF scheduler dominates the resource allocation, but the system fails to sufficiently explore independent RIS states within the averaging horizon T_c, limiting opportunistic gains. An intermediate choice of T_s provides the best balance: it is long enough to support stable transmission under favorable RIS alignment, yet short enough to ensure that multiple RIS selections occur within T_c, thereby preserving fairness and convergence guarantees. It is worth noting that this observation differs from the simulation results, where smaller T_s values appeared optimal due to the idealized environment that omitted control and MCS adaptation delays. In the experimental testbed, these non-ideal factors become dominant, making moderate switching intervals such as $T_s = 5$ s most effective for achieving both high throughput and stable link adaptation.

Table 4.3: Tradeoff between PF averaging window T_c and RIS switching time T_s.

T_c (slots)	**RIS switching time T_s (sec)**				
	1	**3**	**5**	**9**	**15**
200	41.1	39.5	47	47.9	40.3
2000	41	41.6	50.1	48.2	42.9
20000	46.1	46.7	52.1	52.6	44.7

These observations confirm the theoretical insight of Theorem 3.2: for a fixed T_c, the choice of T_s must balance signaling feasibility with the requirement that T_c/T_s be sufficiently large. In practice, intermediate switching intervals (e.g., $T_s = 5$ s in our setup) achieve the best tradeoff between throughput efficiency and fairness.

4.2.b.iv Randomized RIS Can Procure Full Benefits

In this sub-section, we show that a randomized RIS, when paired with an opportunistic scheduler, can achieve nearly the maximal throughput associated with optimized RIS configuration. Fig. 4.7 depicts the system throughput versus T_c, where T_c is the EWMA parameter in (2.7). The topmost curve corresponds to round-robin (RR) scheduling of UEs with the RIS optimally configured toward the scheduled UE in each slot. This genie-aided curve is obtained by averaging the single-UE throughput values with RIS beamforming to the considered UE (cf. Table 4.2); it is independent of T_c and represents the maximum achievable throughput with an optimized RIS. Practically, achieving this throughput requires tight control of the RIS, configuring it based on the scheduled UE. For comparison,

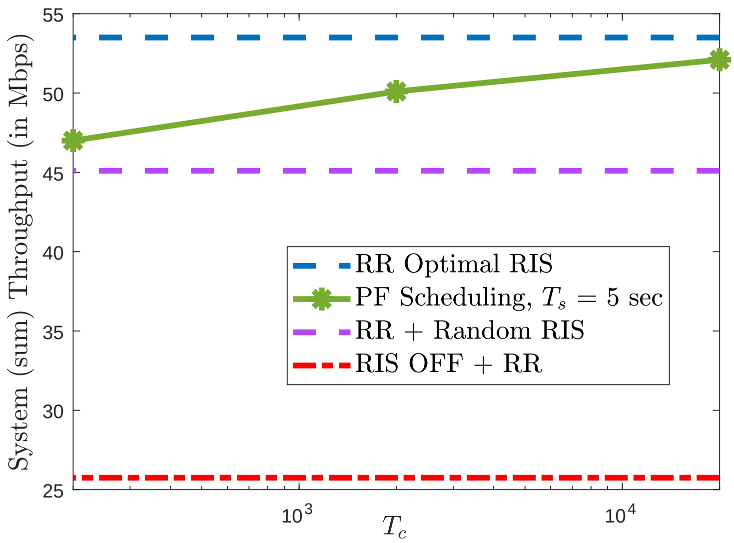


Figure 4.7: System throughput vs. T_c.

we also include the case of RR scheduling with *random* RIS switching. This curve appears below the PF scheduling curve, since RR does not exploit instantaneous channel variations and suffers from the randomness of RIS states. When the RIS switches randomly according to (4.3) and the gNB employs PF scheduling, the system behavior is strongly influenced by the averaging window T_c. For small T_c, the scheduler prioritizes long-term fairness, allocating resources more uniformly across users and thereby reducing instantaneous throughput. As T_c increases, the scheduler becomes more responsive to instantaneous channel fluctuations, tending to select the UE that temporarily benefits from the RIS beam direction. As a result, the PF throughput curve lies above the RR-with-random-RIS baseline and gradually approaches the genie-aided optimized RIS throughput as T_c becomes larger, confirming the opportunistic scheduling gain predicted in [73].

Importantly, the figure demonstrates that the throughput achieved with a randomized RIS and a properly tuned PF scheduler at the gNB not only consistently surpasses that of the no-RIS case, but can also deliver the full benefits of an optimized RIS even without explicitly optimizing the RIS.

To further substantiate the above findings, Fig. 4.8 presents a histogram showing the fraction of time slots in which each UE is scheduled during its respective favorable (good) and unfavorable (bad) states, based on a single scheduling instance with $T_c = 20000$ observed over a long time trace of channel realizations. Specifically, the figure shows the

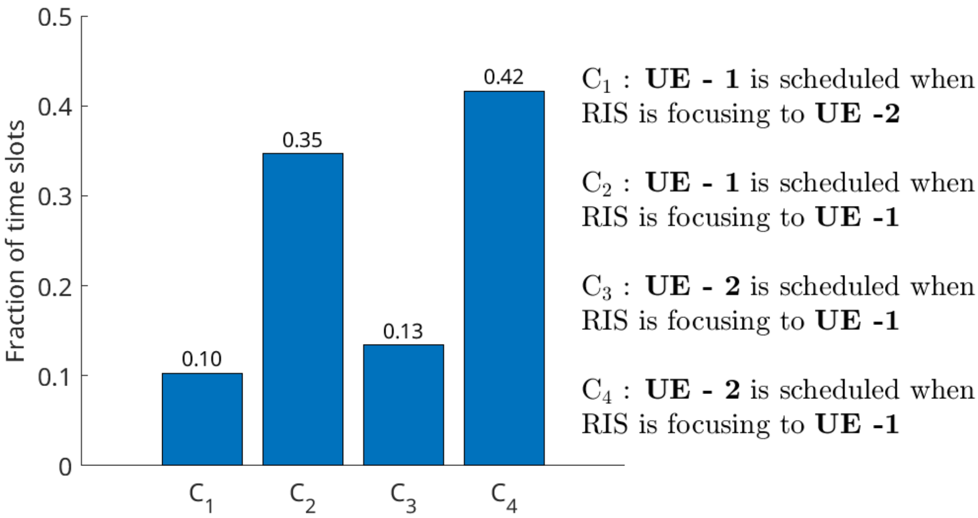


Figure 4.8: Fraction of time slots allotted by PF scheduler with $T_c = 20000$ and $T_s = 5sec.$

proportion of time slots in which a PF scheduler selects a UE for transmission when the RIS is aligned with that UE's channel versus when it is not. We see that, despite the randomized nature of the RIS configurations, the scheduler predominantly transmits data to a UE when the RIS is in the beamforming configuration favoring that UE. Occasional instances occur where a UE is scheduled even when the RIS is not aligned toward it, for example, when packets need to be retransmitted (in 5G NR, retransmissions are prioritized above PF scheduling decisions).

Moreover, the PF scheduler also ensures that the UEs are scheduled almost equally over time, each receiving about 40% of the total time slots during which the RIS focuses its energy towards them, thereby maintaining fairness across UEs. Since the RIS chooses its configurations randomly, the duty cycles of favorable states vary. For illustration, in the first 120 observations (roughly 3 minutes), UE-2 happened to experience more favorable configurations than UE-1, resulting in a larger number of slots allocated to UE-2. This randomness can also be observed in the RSRP plot.

4.2.b.v Randomized RIS Always Helps

To conclude, we demonstrate that even a randomised RIS operation provides consistent throughput gains even when users are not in its beamforming configuration. In this experiment, two UEs are served while the RIS operates with two beamforming directions, 30°

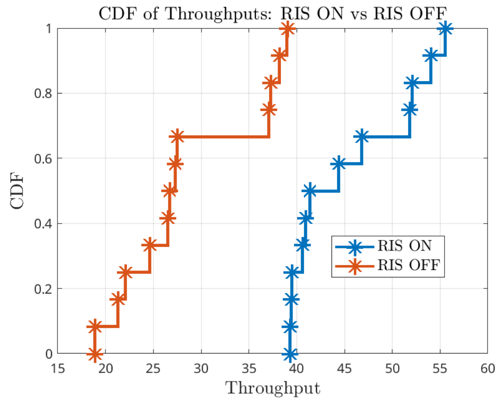


Figure 4.9: Cumulative distribution function (CDF) of measured throughput values with and without RIS.

and 60°. The RIS switches randomly between these two configurations every 5 seconds, and the PF scheduler employs an averaging window of $T_c = 20000$. Measurements were taken across different user placements and distances from the RIS, covering three representative scenarios: (i) both UEs in beamforming configurations, (ii) one UE aligned with the RIS beam while the other is not, and (iii) both UEs outside the beamforming directions. For each scenario, identical measurements were repeated without RIS to provide a baseline. Figure 4.9 shows the cumulative distribution function (CDF) of the measured throughput across these settings. The CDF curves with RIS consistently lie to the right of those without RIS, indicating higher throughput values. This improvement holds not only when both UEs are in favorable beamforming configurations, but also when only one or even neither UE is directly aligned with the RIS beam. The randomized RIS switching ensures that, over time, each UE benefits from favorable states, while the PF scheduler opportunistically exploits these channel variations.

These results confirm that randomized RIS operation always helps: even without tight control of RIS phases, the statistical diversity introduced by random switching improves system throughput relative to the baseline without RIS. In practice, this means that RIS deployment can yield gains even under simple randomized operation, without requiring per-slot optimization of RIS configurations.

5 | Conclusion

This work experimentally demonstrated the performance benefits of integrating a reconfigurable intelligent surface (RIS) with a real-time 5G New Radio system. In addition to confirming that the presence of an RIS significantly improves system performance compared to a baseline setup without RIS, the results show that a randomly configured RIS is capable of achieving a substantial portion of the attainable gains. By exploiting the inherent opportunistic proportional fair scheduling mechanism of 5G NR, which naturally prioritizes user equipments experiencing favorable instantaneous channel conditions, near-optimal performance can be achieved without explicitly optimizing the RIS phase configurations. This enables a low-complexity and practically viable integration of RIS into existing 5G systems. A comprehensive experimental evaluation of key performance metrics, including reference signal received power (RSRP), block error rate (BLER), modulation and coding scheme (MCS) index, and throughput, further validates the effectiveness of randomized RIS operation. In addition, an analytical investigation of the interdependence between the RIS switching interval and the exponential weighted moving average (EWMA) parameter of the scheduler has been carried out, providing deeper insight into their joint impact on system throughput.

5.1 Publications and Demonstrations

1. **IEEE ICC 2026:** Paper accepted - "Practical RIS Gain Without Pain: Randomization and Opportunistic Scheduling in 5G NR."

2. **IEEE ANTS 2025 (IIIT Delhi):** Experimental demonstration accepted and presented.

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